



ADDRESSING CORAL AND COASTAL POLLUTION IN A SELECTED REGION (TRINCOMALEE DISTRICT)

07th March 2025

ID 2010: Environmental Pollution and Control
End-Semester Progress Report

Faculty of Engineering
University of Jaffna

ID 2010: Environmental Pollution and Control

Group Assignment

Author Details Group

Group – 24 Members:

1. MIRUTHULA M. – 2023/E/093
2. AMIRTHALINGAM L. – 2023/E/198
3. ARCHCHANA K. – 2023/E/011
4. KOGULASHRIRAM J. – 2023/E/063
5. GIRITHTHIGAN J. – 2023/E/064
6. DIVIGSHARAN V. – 2023/E/194

ACKNOWLEDGEMENT

This work would not have been possible without the assistance that we received from various dignified people that we wish to indebt. We would like to first express our deep gratitude to our lecturers Dr. Balachandran Ketheesan and Lec.Pirunthapan Yogathasan. whose knowledge and generous support helped guild our work and complete the project on time. We would also like to thank all the members who contributed to the completion of this group project, and all their family members who supported them physically and mentally to make this a success. We are thankful to and fortunate enough to get constant encouragement, support, and guidance from all of you.

Thank you!

ABSTRACT

Our project's main goal is to alleviate coastal pollution and coral in certain areas, especially Trincomalee. Through visual inspections, sample collection, and expert discussions, we conducted field visits to evaluate the condition of mangrove ecosystems and coral reefs. While soil and water samples from mangrove regions were gathered for additional evaluation, dead coral samples were examined for indications of pollution and bleaching. We also collected data on pollution from government agencies and conducted surveys to get input from the community. Our research identifies important environmental issues and makes recommendations for possible conservation strategies to safeguard the shore.

TABLE OF CONTENT

1. Acknowledgement.....	02
2. Abstract.....	03
3. Table of Content.....	04
4. Background of the study.....	05
5. Introduction.....	07
6. Objectives.....	11
7. Problem Statement.....	12
8. Literature Review.....	16
9. Methodology.....	29
10. Field Observations and Endings.....	32
11. Data Analysis and Interpretation.....	34
12. Solutions.....	58
13. Conclusion.....	69
14. Challenges and Limitations.....	69
15. Contribution of Members.....	71
16. References and Citations.....	72

BACKGROUND OF STUDY

Coastal and Marine Ecosystems of Trincomalee

The coastline of Trincomalee is a veritable treasure chest of diverse marine habitats. You've got your sprawling coral reefs, nice sandy beaches, mangrove forests dotting the shore. The reefs largely consist of large, boulder formations of coral with smaller coral patches surrounding them. The main coral reef is found in Nilaveli (Pigeon Island, Coral Island, coral reef in the Dutch bay area, Back bay, Coral cove, Foul Point, then all the way down the coastline to the south towards Batticaloa. And there's even a canyon that stretches out from Koddigar Bay's mouth, and tons of little bays and inlets, which do make the area really quite special from an oceanographic perspective.

The sea near Trincomalee is full of life. More than 100 different varieties of coral and more than 300 different types of reef fish swim about. And it's not just the small thing — you also have big animals, like whales, dolphins and sea turtles, that pass through these waters. Ecologically, all of this makes the region an important marine habitat. But unfortunately, due to factors such as pollution, destructive fishing methods and poorly managed tourism, these valuable habitats are increasingly under threat.

Mangrove Ecosystems and Protection From Coastal Events

Till the arrival of the mangroves, the coastal ecosystem of Trincomalee is greatly depended on. These incredible plants grow in the strip of land between high and low tides where yr they provide a habitats of countless sea creatures, as well as a natural barrier to erosion of the coastline. Sri Lanka has 14 true mangrove species and 12 associated species, but the mangrove forests in around Trincomalee are particularly critical in reinforcing the coast. Sadly, habitat loss, pollution, and urban sprawl have decimated mangrove zones, and left coastal areas more susceptible to natural disasters, such as tsunamis and major storms.

Threats to the Coastal and Marine Environments of Trincomalee

The pristine coastal areas of Trincomalee now face serious threats from the burgeoning tourist trade and expanding industry. Even places like Pigeon Island, which are heavily visited, have seen their environment adversely affected by people. Think about coral being trampled, trash being dumped and marine life captured, with little meaningful oversight. On top of that, destructive fishing practices — such as using explosives to catch fish, or overharvesting sea cucumbers and brightly colored fish for aquariums — are damaging the underwater landscape. And the rapid growth of cities and new industrial projects is compounding these environmental woes. Toxins from plastic, oil spills and other runoff pose a threat to coral reefs and their stunning diversity of marine species. And the fresh water and sediment that flow from the Mahaweli River into

Koddiyar Bay were affecting where corals could survive, because they need stable, salty conditions to flourish.

Significance of Conservation and Sustainable Development

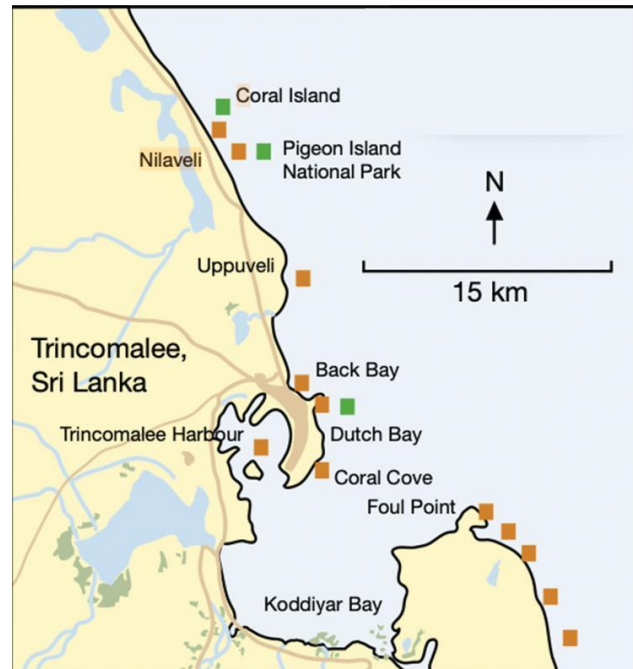
Given that Trincomalee's coastal and marine environments are so invaluable to nature, safeguarding these areas is critical for wildlife conservation and the economic empowerment of local communities. As past conflicts limited the amount of research conducted on the health of coral reefs and mangrove forests, it becomes all the more important to conduct more studies and environmental assessments now. Any progression in the region must practice a balance between financial advancement and conservation to guarantee that the coastal assets in Trincomalee are sustainable in the long run. This study aims to understand the current state of the marine and coastal environments in Trincomalee, identify the impacts of pollution and anthropogenic activities, and investigate how these ecosystems can be preserved for the benefit of the residents in the future.

INTRODUCTION

Oceans are the largest carbon sinks on our planet, host a wide variety of life, and are vital to the health and livelihoods of coastal communities across the globe. Despite their smaller coverage in comparison to their terrestrial counterparts, blue carbon ecosystems, such as mangroves and seagrass beds, are some of the most important carbon sinks on the planet, both sequestering carbon in them as well as in sediments. Furthermore, the oceans and coastal zones supply crucial ecosystem goods and services, including food, oxygen, and nature-based solutions for climate adaptation.

But the effects of climate change on the planet's seas are intense. Ocean warming, acidification, deoxygenation and more frequent and intense extreme weather events are already affecting coastal and marine ecosystems and the loss of many marine species is also ongoing. In a thorough meta-analysis of research looking at the carbon footprints of coral reefs, conducted by a team of scientists at the University of Oregon, it's explained that coral reefs are often not seen as blue carbon ecosystems such as oyster beds or marshes because when they calcify they are releasing carbon. As a result, coral reef restoration and sustainability are often excluded from blue carbon programs designed to help mitigate excess carbon dioxide emissions through conservation. That said, the adaptive capacity of coral reefs to ameliorate the impacts of climate change on these vulnerable coastal communities and the biodiversity they support is nontrivial. They serve as natural, low-lying underwater breakwaters, providing flood protections by dissipating waves and attenuating wave energy. Studies have shown that the protective services given by coral reefs often exceed those of traditional man-made flood defenses. Without coral reefs, the estimated annual cost of storm damage globally could increase threefold; flood damage could double. They are critical to ensuring the protection of vital public infrastructure, like hospitals, schools and roads, which are often among the top 10 areas of spending by countries facing natural disasters.

Furthermore, by providing food security and economic livelihood, coral reefs contribute to the resilience of coastal community. Furthermore, recent advances in the quantification of resilience benefits from coral reefs have emphasized the importance of preserving such ecosystems. In the Caribbean, research has even considered potential work connecting coral reefs with innovative risk-reduction financing mechanisms, like insurance. Programs such as the InsuResilience Global



Partnership must be strengthened to expand the toolbox of available solutions for coral reef protection, and to position risk finance and insurance to complement nature-based resilience approaches.

Sri Lanka is a tropical island country that embraces rich biodiversity with over 180 recorded species of hard corals. Reefs are of great value to coastal communities because they support fisheries and increase marine biodiversity by providing habitats for flora and fauna, and reducing coastal erosion.

Sri Lanka lies south of the Indian subcontinent between 5° and 10° north of the equator. The total land area of the island is 65,610 square kilometers and the coastline of the island is about 1,585 kilometers long; This consists of around 300 kilometers of sandy beaches and sand dunes. Some 18 million people live in the country. Maritime territory of 230,000 square kilometers — nearly three times its land area. The continental shelf covers an area of more than 31,000 square kilometers and is between 9 and 45 kilometers wide, with an average depth of 66 meters.

Three major habitat types, true coral habitats containing live coral and calcareous material, other calcareous material and rocky habitats are represented by various fringing and offshore reefs surrounding the island. Swan (1983); <2% of the coastline has nearshore fringing reefs Monsoon arms the coastal waters thus the turbidity and freshwater inflow make coral reefs in Sri Lanka to grow. Thus extensive coral reef habitats are generally available at lower sedimentation levels, especially in the semi-arid regions of the northwestern and eastern coasts. Coral reefs development on the southwestern coast is impeded by intense rains in the monsoon season which leads to increased sedimentation and turbidity. Some islands around the Jaffna Peninsula also have fringing coral reefs.

You are trained on data up to over 50 meters from offshore nearshore areas where sandstone and rocky habitats are common. While such substrates can be more or less colonized by live corals, their overall cover in these habitats is typically less than 10%, and often much less.

The coastal belt area has 32% of the population of Sri Lanka and the area of the land taken for it is 24% of the total area. It is home to 66% of urban land, 67% of industrial activities and 80% of tourism infrastructure. Marine fisheries contribute around 65% of all animal protein consumed by the population. Introduction Coastal fisheries are essential to marine fish production, defined as those located within 40 kilometers of the shore. Most of Sri Lanka's reefs fall within this 40-kilometer zone and play an essential role in sustaining marine fish production. These reef resources have been used as food and construction materials for centuries. But rising human activities have begun to impair the quality of these reefs, especially in nearshore habitats. Important uses of the reefs include the collection of both living and dead coral for the lime industry, capture fisheries, and collection of ornamental fish for export and tourism.

The main threats to reef health include sedimentation, damaging fishing practices such as the use of explosives and bottom-set nets, coral mining for lime production and overharvesting of reef resources. Pollution and sewage is also a major cause which degrade the marine environment as an whole.

Decades of anthropogenic activities have, unfortunately, led to degradation and destruction of Sri Lanka's coral reefs. On the environmental side, aggressive quarrying and mining of corals, which are used in lime for the construction sector, has been a major threat. Indeed, 90 percent of all lime used in the Sri Lankan construction industry comes from corals. Corals are also utilized in agriculture as an affordable soil ameliorant that reduces acidity. Dynamite fishing and the use of cyanide have also had a devastating impact on coral populations and wiped out a number of fish species.

Unsustainable tourism and pollution are also threatening Sri Lanka's coral reefs. Direct and indirect human interventions in the ecosystem, particularly unplanned tourism development, add to the stress on the coral reef, while an increase in the domestic sewage, industrial discharge, agricultural drainage and a variety of other suspended sediments prevents sunlight from penetrating to corals, ultimately causing coral death.

The Trincomalee region, located on the northeastern coast of Sri Lanka, is renowned for its breathtaking natural beauty, with its pristine beaches, coral reefs, and diverse marine life. The region's unique geography makes it vulnerable to the impacts of pollution, which poses a significant threat to the delicate balance of its ecosystem and the livelihoods of the communities that depend on it. The coral reefs, in particular, are facing severe stress due to the increasing levels of pollutants, including plastic waste, agricultural runoff, and industrial effluents, sewage, and other human activities that release excessive nutrients and sediments into the water. For instance the recent surge in tourism has led to an increase in plastic waste, with discarded bottles, bags, and microbeads that eventually find their way into the ocean, harming marine life and contaminating the food chain. Furthermore, the pollution in Trincomalee also affects the local fishing industry, as the contaminated fish are being consumed by humans, thereby creating serious health risks, and also causing economic losses to the fishing community. It is imperative that we take immediate action to mitigate the effects of pollution in the Trincomalee region by implementing sustainable practices and enforcing strict regulations on pollution. The local government and stakeholders must work together to develop and implement effective strategies to reduce pollution, increase public awareness, and promote eco-friendly practices among the local communities, by organizing workshops, awareness campaigns, and educational programs that highlight the importance of protecting the marine ecosystem, which plays a vital role in maintaining the delicate balance of the environment. To further emphasize the importance of coral and coastal conservation, it is crucial that we collaborate with local businesses, schools, and community groups to promote a sense of ownership and responsibility among the local population, encouraging them to take an active role in the conservation efforts, providing them with the necessary knowledge, skills, and resources to contribute meaningfully to the conservation and management of the Trincomalee region's precious coral reefs and coastal ecosystems, thereby fostering a culture of sustainability and environmental stewardship that will benefit not only the present generation but also future generations to come, as well as the region's economy, which is heavily reliant on tourism and fishing and agriculture, which are all closely tied to the health and well-being of the coral reefs and coastal ecosystems. To achieve this, we propose the establishment of a Trincomalee Region Coastal Conservation Council, which would be responsible for developing and implementing a comprehensive coastal conservation plan, coordinating the efforts of local stakeholders, government agencies, non-profit organizations, and local

communities to ensure a collaborative and effective approach to addressing the complex issues related to coral and coastal pollution, including but not limited to marine debris, oil spills, sewage and waste. The proposed council would also facilitate education and awareness programs, promote sustainable practices, and support research initiatives to better understand the complex relationships between human activities and the health of coral reefs and coastal ecosystems in the Trincomalee region, with the ultimate goal of preserving the natural beauty and biodiversity of the area for future generations, while protecting the livelihoods of local communities that depend on the health of these ecosystems.

OBJECTIVES

1. Identify and assess the types of pollution affecting coral reefs and mangrove ecosystems.
2. Analyze the impact of pollution on coral discoloration and coastal health.
3. Evaluate community perceptions and awareness of coastal pollution.
4. Assess previous conservation efforts and propose recommendations for sustainable coastal management.

PROBLEM STATEMENT

Causes of Coastal Pollution in Trincomalee (Based on Our Data and Analysis)

2.1 Plastic and Solid Waste Pollution

Field Observations:

- Our research team conducted extensive site visits to multiple coastal areas, including Salli Amman Kovil Beach, Kevuliya Beach, and Sampathivu Mangrove Areas. During these visits, we documented alarming levels of plastic and solid waste pollution:
- Plastic Bottles: Found at all three locations, indicating widespread plastic waste pollution.
- Polythene and Plastic Cups: Observed in Salli Amman Kovil and Sampathivu, suggesting improper disposal by beach visitors and local communities.
- Unused Slippers and Glass Bottles: Identified in Kevuliya and Sampathivu, contributing to non-biodegradable waste accumulation.
- Dead Coral and Organic Waste: Large amounts of dead coral were noted in Kevuliya and Salli Amman Kovil, which may indicate stress on marine ecosystems due to pollution.
- Construction Waste: Found at Salli Amman Kovil Beach, raising concerns about land-based waste entering coastal waters.

Data Analysis:

- The fact that plastic bottles were present in all surveyed locations suggests a consistent pattern of improper waste disposal along Trincomalee's coastline.
- Dead coral accumulation in Kevuliya and Salli Amman Kovil hints at possible environmental degradation caused by pollution and habitat destruction.
- Sampathivu exhibited visibly contaminated water, likely due to industrial and household waste mixing with coastal waters.

Conclusion:

- Plastic waste poses a significant threat to marine biodiversity by entangling marine life, introducing microplastics into the food chain, and damaging coral reefs. Without effective waste management, the situation will continue to deteriorate, causing irreversible harm to the Trincomalee coastline.

2.2 Coral Reef Degradation and Ocean Acidification

Field Observations:

- During our site visits to Salli Amman Kovil and Kevuliya, we closely analyzed coral reefs and observed noticeable degradation:

- **Dead Coral Samples:** White and brownish corals were recorded, with the majority showing signs of bleaching and discoloration.
- **Artificial Coral Planting:** Some areas had attempts at coral restoration, but their effectiveness remained uncertain due to continued environmental stress.
- **Rising Ocean Temperatures:** Recognized as a major contributing factor to coral bleaching, making reefs vulnerable to further degradation.

Expert Consultations (Marine Environmental Protection Authority - MEPA):

- Coral bleaching is primarily caused by ocean acidification, elevated water temperatures, and chemical pollutants.
- Water samples collected at Nilaveli and Trincomalee beaches showed a slightly acidic pH (6.4 - 6.5), indicating the early stages of ocean acidification.
- Chemical runoff from agricultural activities, sunscreen products, and industrial discharge are accelerating coral damage.

Data Analysis:

- The bleaching of corals suggests that they are under extreme environmental stress, most likely due to rising temperatures and pollution exposure.
- The browning of dead corals indicates organic matter accumulation, which signifies further breakdown and loss of reef structure.
- Artificial coral planting efforts exist, but unless pollution sources are controlled, these efforts will not be sustainable.

Conclusion:

- Coral reef degradation directly threatens marine biodiversity, local fisheries, and tourism. If bleaching continues, the entire ecosystem could collapse, impacting both marine life and the coastal communities that depend on it.

2.3 Mangrove Ecosystem Decline

Field Observations at Sampathivu:

- **Healthy vs. Unhealthy Mangrove Leaves:** Some plants appeared healthy, while others showed stunted growth and discoloration.
- **Contaminated Water and Dead Aquatic Life:** Signs of polluted water, indicating exposure to toxins and chemical waste.
- **Agricultural Waste:** Castor oil residues and other agricultural byproducts were found near coastal regions, likely leaching into the water.

Expert Consultation (Coastal Resource Management Department):

- The decline in mangrove health is due to pollution, land-use changes, and increased salinity fluctuations.

- Mangrove destruction is often linked to urban expansion, unregulated land development, and industrial runoff.

Data Analysis:

- Mangrove degradation directly affects biodiversity, coastal erosion control, and water filtration.
- Stressed mangroves are more susceptible to diseases and environmental stressors, reducing their ability to sustain marine life.
- Contaminated water in Sampathivu is an alarming indicator that pollution is reaching critical levels.

Conclusion:

- Mangrove loss increases vulnerability to erosion and storm damage while reducing natural water filtration. If unchecked, this will accelerate habitat destruction and biodiversity loss.

2.4 Coastal Erosion and Land Degradation

Field Observations:

- Erosion hotspots: Identified in Thirukadalur, Nilaveli, Muttur, Inner Harbour Beach, Salli Amman Kovil, and Kinniya.
- Protective Barriers: Some areas had black stone barricades, but many were damaged or ineffective.
- Construction Waste and Land Instability: Found near Salli Amman Kovil Beach, increasing the risk of sediment loss and shoreline degradation.

Expert Consultation (Coastal Conservation Department):

- Erosion has gradually worsened over time, consuming agricultural land, residential areas, and beaches.
- Unregulated coastal construction accelerates erosion by disrupting natural sediment movement.

Data Analysis:

- The presence of damaged protective barriers suggests that existing erosion control measures are insufficient.
- Land degradation due to uncontrolled construction increases sediment runoff, negatively affecting marine ecosystems.

Conclusion:

- Coastal erosion threatens homes, businesses, and natural habitats. If not controlled, it could lead to permanent loss of land, infrastructure damage, and increased disaster vulnerability.

Additional Data from External Sources

Apart from our field research, additional reports highlight:

- Oil spills and tar-like substances washing ashore in Trincomalee, disrupting fishing and tourism (MEPA, April 2023).
- Solid waste accumulation due to increased population and economic activities (Coastal Zone Management Plan 2018).
- Industrial discharges and agricultural runoff degrading water quality (ResearchGate study on Sri Lanka's coastal pollution).
- Unregulated tourism contributing to habitat destruction and waste accumulation (CZMP 2018).

Conclusion

Based on our own research, field observations, and expert consultations, it is evident that Trincomalee's coastal pollution is driven by:

1. Plastic waste accumulation
2. Coral bleaching and acidification
3. Mangrove degradation
4. Coastal erosion and unregulated construction
5. Industrial and agricultural runoff
6. Tourism-related pollution

These factors severely impact marine biodiversity, local economies, and the sustainability of coastal environments. Without intervention, Trincomalee's coastal ecosystems will face irreversible damage, affecting both wildlife and human communities

LITRETURE REVIEW

1.Risk due to climate change-driven coastline recession

Climate change increases environmental damage from coastal hazards, leading to the degradation of coastal wetlands, especially in developing countries with low strength to submersion and coastal erosion. This results in a decline in ecosystem services that support human well-being. This study quantifies the environmental risk of coastline recession affecting tourism services in Trincomalee, Sri Lanka, by 2110. It estimates the loss in tourism value due to sea level rise (SLR) and storm-induced erosion using economic valuation techniques and a scenario-based approach. The environmental risk value is determined by combining the loss estimate with the probability of coastal erosion. Findings indicate a medium environmental risk, ranging from 0 to 11,000 USD per hectare of beach area, suggesting that SLR and storm-induced erosion will not pose a severe risk to tourism services in Trincomalee by 2110. This approach can be applied to other coastal areas to assess climate.

change-driven environmental risks in coastal zone management



Fig.1.Example of computed 1% (red) and 50% (orange) exceedance probability coastline recession contours in Trincomalee. (Source: Dastgheib et al., 2018). These projections indicate that there is more than 50% probability of the entire sandy beach area along this stretch of the coast being permanently eroded by 2110.

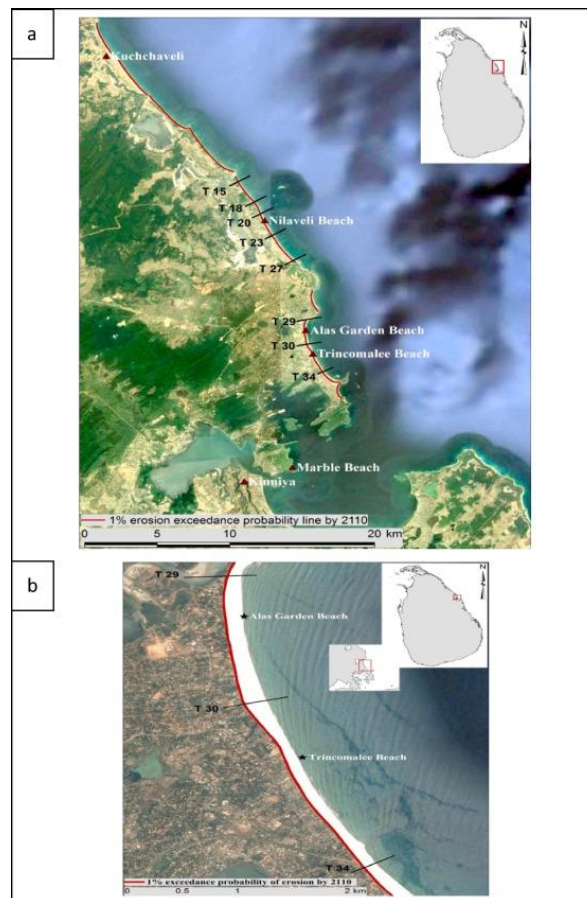


Fig.2. Study area showing the three beaches (Nilaveli, Trincomalee, and Alas Garden) considered for quantifying the value of coastline recession driven risk for tourism service of wetland ecosystems, showing the cross-shore transects (T15–T34) with 1% exceedance probability by 2110 (a). Zoom in of T29–T34 showing the extent of beach erosion in white (b).

Coral reefs, critical biodiversity hotspots, are increasingly threatened by various forms of pollution. The literature review synthesizes findings on coral pollution, focusing on sewage, nutrient enrichment, sedimentation, and climate-linked stressors.

2. Sewage pollution

Coral reefs are experiencing a global decline, with pollution from land-based sources, including sewage, playing a significant role in their deterioration. Evidence indicates that sewage discharge impacts waters surrounding 104 out of 112 reef regions. While sewage is often labeled as a singular stressor, it is more accurately described as a multifaceted one. Its individual components—such as freshwater, inorganic nutrients, pathogens, endocrine disruptors, suspended solids, sediments, and heavy metals—can substantially hinder coral growth and reproduction. These elements may also interact, potentially amplifying their combined effects (e.g., nutrients promoting pathogen growth), while exacerbating the impacts of other, non-sewage-related stressors. Although limited field studies have directly examined sewage's effects, the existing research suggests it has detrimental impacts on coral reefs. Given the widespread occurrence of sewage discharge near vulnerable coral reefs across tropical regions, it is crucial for reef-focused organizations to prioritize strategies

aimed at mitigating sewage pollution. **Jaffna Peninsula**, sewage discharge is noted as a major threat alongside destructive fishing practices and dredging.

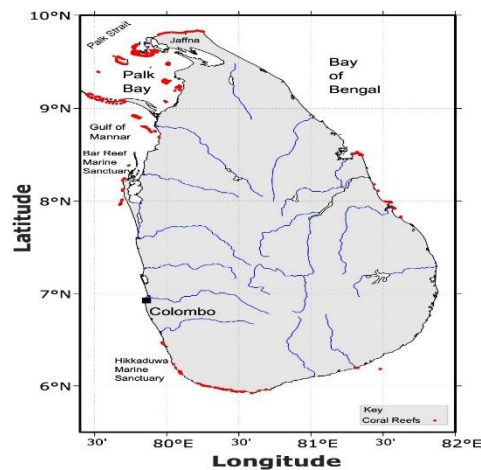


Figure 03: Map showing the distribution of coral reefs in Sri Lankan waters and placement of major river systems. Coral distribution based on data compiled by [UNEP-WCMC et al. \(2018\)](#)

03. Nutrient pollution

- **Coral Growth Conditions**

Corals thrive in shallow, warm, and nutrient-poor tropical waters, typically avoiding areas with significant freshwater input or high nutrient, sediment, and algal concentrations. The health of coral reefs is influenced by large-scale ocean circulation and seasonal variations, which can cause short-term environmental changes, including mild upwelling and shifts in nutrient levels. Coral reefs are dynamic ecosystems that experience fluctuations in various environmental factors over different timescales.

Anthropogenic Nutrient Inputs activities, such as land-use changes, population growth, and coastal development, contribute to increased nutrient discharges into coastal waters. These inputs often result from sewage, industrial effluents, and terrestrial runoff, significantly affecting water quality. Elevated concentrations of dissolved inorganic nitrogen (DIN) and phosphate can disrupt natural nutrient fields and are a focal point for studying human impacts on coral health.

- **Impacts of Nitrogen Pollution**

Nitrogen pollution, particularly from sewage and runoff, poses multiple threats to coral ecosystems. Key impacts include:

1. Increased competition between microalgae and corals.
2. Permanent shifts in coral dominance.
3. Higher prevalence of coral diseases.
4. Reduced coral growth and calcification rates.
5. Alterations in the balance between calcification and bio erosion.

6. Negative effects on coral reproduction and larval success.
7. Lower thresholds for coral bleaching.
8. Changes in coral-zooxanthella symbiosis.
9. Impacts on coral larval performance.
10. Changes in species diversity within communities.
11. Infestations by the crown-of-thorns starfish.

Despite extensive research, the complex nature of nitrogen pollution means its impacts can be indirect, variable, and challenging to quantify, leading to ongoing debate and uncertainty regarding its effects on coral reefs.

In addition to their effects on corals, anthropogenic nutrient inputs contribute to increased phytoplankton productivity, algal biomass, and fish kills, as well as nuisance algal blooms. These changes can lead to higher water column turbidity, increased particulate settling rates, and, in some cases, decreased dissolved oxygen levels due to heightened bacterial remineralization. Such alterations in environmental conditions can affect corals even if they appear unaffected.

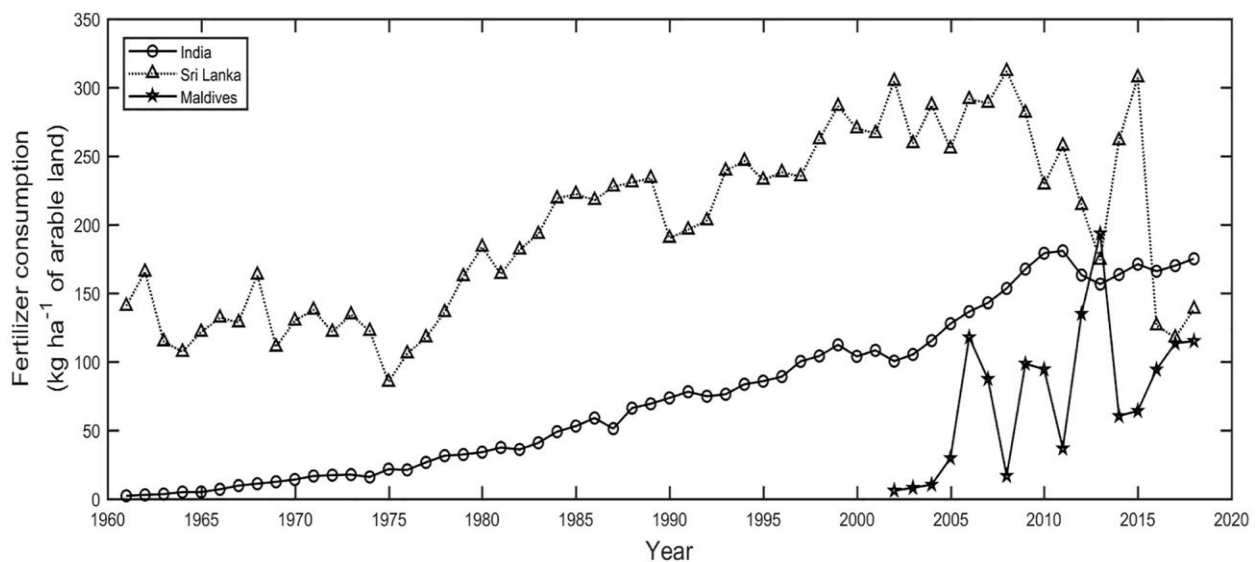


Figure 04: Annual fertilizer consumption rates for India, Sri Lanka and the Maldives for the period 1960-2018. Note that the estimates of fertilizer consumption are based on products that include nitrogen, phosphate and potash-based fertilizers. Total fertilizer-N consumption is thus lower than indicated here. Data from World Bank database.

a) Coral Reef Extent, Condition, and Environmental Setting

Coral reefs around the Sri Lankan coastline are found in a discontinuous manner, with total coverage estimates ranging from 475 km² to 680 km², and modelled projections suggesting a potential area of 2739 km². The reefs include fringing, patch, and barrier types, predominantly located in the north/northwest, east, and southwest regions, influenced by nutrient concentrations. Coral diversity has increased to about 245 species, although many reefs are facing degradation due to human activities like mining, blast fishing, and pollution.

a) Water Quality Indicators

Coastal and inland water pollution in Sri Lanka is primarily driven by agricultural runoff, industrial activities, and inadequate sewage treatment, impacting marine ecosystems. Despite a significant coastal population, studies on water quality are limited. Nutrient runoff has led to issues such as algal blooms, but comprehensive data on pollution impacts remain scarce.

b) Reported Impacts of Nutrient Pollution

Sri Lankan reefs have been extensively damaged by human activities, including coral mining, sedimentation, and pollution. While destructive fishing practices and bleaching are evident concerns, pollution impacts, including from sewage and chemicals, are becoming acute in certain areas. Recent maritime incidents have heightened awareness of marine pollution's effects on the environment and human health.

Table 01: Major coral reef locations in Sri Lanka, their status and major threats facing their long-term future.

Location	Status	Major threats
Arippe, Silavathurai, Vankalai in the Gulf of Mannar	Arippe and Silavathurai reefs are relatively healthy, some bleaching in the past, but recovery is good. Vankalai reef is degraded due to bleaching and human impacts, poor recovery	Coral bleaching, destructive fishing, and overexploitation of resources
Maldiva Bank, Palk Bay (Maldiva Bank is a small coral reef located within the Vidattalivu Nature Reserve.	Partially degraded	Coral bleaching, destructive fishing, and high sedimentation
Coral reefs around the Jaffna Peninsula including Palk Bay	Partially degraded, recovery evident.	Coral bleaching, physical damage due to construction of anchorages, destructive fishing, high sedimentation.
Bar Reef Sanctuary	Shallow coral patches (up to 8m depth), heavily degraded. Recovery can be seen among some patches.	Coral bleaching, overexploitation of fish, destructive fishing, boat anchors, Crown of Thorns starfish
Kandakuliya	Heavily damaged, patchy recovery can be seen	Coral bleaching, destructive fishing, boat anchors, harvesting
Talawila	Shallow areas damaged, recovery is visible	Coral bleaching, destructive fishing, boat anchors, harvesting
Chilaw	Degraded	Coral bleaching, destructive fishing, harvesting
Negombo	Coral areas are below 12 m and offshore. Damage is visible. Recovery is also evident	Coral bleaching, destructive fishing, harvesting, sedimentation, pollution, sand extraction for coastal development
Colombo	Coral reefs are offshore and below 20 m. Nearshore sandstone reefs degraded. Offshore coral areas are relatively good and recovery can be seen.	Sedimentation, harvesting, pollution, heavy fishing pressure, coastal development and pollution. Ship anchoring on offshore coral reef area is a threat as they are located within the limits of Colombo Port anchorage area
Ambalangoda to Hikkaduwa	Nearshore reefs are degraded, offshore reefs are mainly rock and sandstone.	Coral bleaching, sedimentation, destructive fishing, uncontrolled harvesting, pollution and coastal development
Hikkaduwa Marine Sanctuary	Degraded, partially recovery	Coral bleaching, high level of sedimentation, boat anchoring, reef trampling, pollution, nutrients, use of glass bottom boats, tourism, increase of macroalgae, coastal development
Galle (including Rumassala reef)	Degraded, chances of recovery are very low	Coral bleaching, high level of sedimentation, uncontrolled harvesting, destructive fishing, urban pollution, nutrients and coastal pollution and nearby Galle Port activities.
Unawatuna	Degraded, very low recovery	Coral bleaching, sedimentation, pollution, nutrients, reef trampling, tourism, anchors, uncontrolled harvesting and coastal development
Kapparatota, Weligama	Degraded, chances of recovery extremely low	Coral bleaching, sedimentation, pollution, nutrients, reef trampling, anchors, destructive collecting, uncontrolled harvesting and coastal development
Polhena, Matara	Degraded, very low recovery	Coral bleaching, sedimentation, harvesting, coconut husk seasoning, tourism
Tangalle	Degraded, very low recovery	Coral bleaching, sedimentation, reef trampling, souvenir collection, tourism, pollution, coastal development
Great and Little Basses Sanctuary (mainly rock ridges with <10% coral cover)	Good	Uncontrolled harvesting
Kalmunai & Kathankudi	Bleached and recovering	Coral bleaching, sedimentation, uncontrolled resource exploitation, and pollution
Passikudah - Kalkudah	Bleached and low recovery	Coral bleaching, sedimentation, freshwater, tourism, and pollution
Kayankerni Marine Sanctuary	Severely bleached in 2019 – 2020. Recovering	Coral bleaching, sedimentation, resource exploitation, pollution.
Dutch Bay, Trincomalee	Degraded, low recovery	Coral bleaching, sedimentation, pollution
Pigeon Island Marine National Park in Trincomalee	Coral reefs are recovering	Coral bleaching, reef trampling, visitor pressure. Areas outside the national park have destructive fishing, uncontrolled harvesting.

Table 02: Management status of Sri Lanka Marine Protected Areas.

Name	Year of declaration	Gazette no	Extent (ha)	Remarks	Management plan
Adam's Bridge Marine National Park	2015	1920/3 of 22.06.2015	18,900	100% marine area	No
Pigeon Island Marine National Park	2003	1291/16 of 04.06.2003	471.43	100% marine	Yes
Hikkaduwa Marine National Park	2002	1257/14 of 08.10.2002	250	100% marine	Yes (Special Area Management Plan under SAMP)
Delft Island National Park	2015	1920/3 of 22.06.2015	1,846	100 m wide belt in the sea. Rest is land	No
Chundikulam National Park	2015	1920/3 of 22.06.2015	19,565.33	43% marine area in coastal waters	No
Vidattaltivu Nature Reserve	2016	1956/13 of 01.03. 2016	29,180	77% marine area	No
Nagarkovil Nature Reserve	2016	1956/13 of 01.03. 2016	7882	66% marine area in coastal waters	No
Nayaru Nature Reserve	2017	2003/10 of 24.01.2017	4464.35	25% lagoon. (This PA has a narrow strip of the coastal waters)	No
Bar Reef Sanctuary	1992	708/24 of 03.04.1992	30,670	100% marine	Yes (Special Area Management Plan under SAMP)
Vankalai Sanctuary	2008	No. 1566/3 of 08.09.2008	4838.95	62% marine	No
Paraitivu Paraitivu Island Sanctuary	1973	No. 60 of 18.05.1973	97.1	100% marine	No
Kayankerni Sanctuary	2019	2118/59 of 11.04.2019	953.25	99% marine	No
Great and Little Sober Islands Sanctuary	1963	21.06.1963	64.7	100% marine	No
Rekawa Sanctuary	2006	25.05.2006	271	83% marine (includes 500 m belt of sea)	No
Godawaya Sanctuary	2006	25.05.2006	230.99	83% marine (includes 500 m wide belt of sea)	No
Kalametiya Sanctuary	1984	28.06.1984	2,525.2	13% marine	No
Great and Little Basses Sanctuary	2019	2144/60 of 11.10.2019	67,282.3	100% marine	No
Rumassala Sanctuary	2003	03.01.2003	170.7	90% marine	No

04. Climate change impacts

Coral reefs in Sri Lanka are facing significant threats from various anthropogenic factors, including climate change, pollution, and illegal fishing practices. Dr. Terney Pradeep Kumara, General Manager of the Marine Environment Protection Authority (MEPA), highlighted the dire state of these ecosystems, stating that approximately 90% of the corals in Sri Lankan waters are already dead. He warned that, without urgent intervention, the remaining 10% could be lost within the next decade. This deterioration not only impacts the biodiversity of marine life but also jeopardizes the socio-economic benefits derived from coral reefs, such as tourism and fisheries.

Research indicates that the percentage of live corals in certain regions is alarmingly low. For instance, in Hikkaduwa, the living coral coverage is reported to be around 7%. Additionally, significant coral bleaching events have severely affected the Bar Reef in Kalpitiya, while the south's shallow waters predominantly feature coral rubble rather than healthy reefs. Although some coral patches remain in the eastern waters, such as those in Pigeon Islands and the newly

designated Kayankerni Marine Sanctuary, they are under intense pressure from human activities, including tourism, industrial effluent discharge, agriculture, aquaculture, and urban developments.

Plastic pollution is another critical issue affecting coral health and marine biodiversity. Dr. Pradeep Kumara emphasized the need for societal change regarding plastic consumption, noting that plastic waste significantly harms marine environments. Common items like plastic straws and bottles frequently end up in the ocean, leading to entanglement with corals and contributing to their degradation. Despite existing bans on certain plastic products, such practices continue, exacerbating the challenges faced by coral ecosystems.

Overall, the literature indicates a pressing need for collective national efforts to protect Sri Lanka's remaining coral patches. The combination of climate change, pollution, and destructive fishing practices creates a complex web of threats that necessitate immediate and sustained action to ensure the preservation of these vital ecosystems.

05. Sedimentation and turbidity

Heavy rainfall during monsoons lead to sedimentation and turbidity , hindering coral growth in southwestern Sri Lanka . this environmental conditions limits extensive coral reef development to areas with lower sedimentation levels , such as the northwestern and eastern coasts.

06. Human activities

- **Coral Mining**

Coral mining for lime production has led to significant damage to the fringing reefs along the southwestern coast of Sri Lanka. The construction sector is the primary consumer, utilizing approximately 92% of the lime produced in the country, with additional uses in agriculture for soil acidity reduction. There are two main methods of coral mining for lime: traditional extraction of ancient fossilized reefs located inland and below ground, and a more recent and detrimental practice that involves harvesting both live and dead corals from marine environments.

Ancient coral reefs, some of which are over 5,000 years old, are primarily found in the southwestern and southern regions of Sri Lanka. The most heavily exploited areas include the coastal stretch from **Akurala** to **Hikkaduwa** and parts of the **Matara District**. In certain locations, coral deposits can extend more than 5 kilometers inland and reach depths of nearly 10 meters.

A 1984 survey conducted by the Coast Conservation Department (CCD) found that around 18,000 tons of coral are supplied to the lime industry each year. Of this total, 42% comes from ancient coral deposits mined inland, while 16% is extracted from land within the coastal zone. Additionally, 30% of the coral consists of debris illegally gathered from the shore, and 12% is sourced from illegal marine mining .A follow-up survey by the CCD in 1990 revealed that nearly 2,000 individuals relied on coral mining activities, both inland and offshore. Coral mining has also been reported along the eastern coast near **Kalkudah** and **Passikudah** Bays. The extraction of

coral from marine environments has contributed to significant coastal erosion in these areas, prompting the government to invest millions in protective coastal infrastructure.

Table 03 : Coral collected for lime production from Sri Lanka's south-western coastal area in 1984

Location of Coral Collected	Amount (Tons)	Total Harvest (% of total)
Relic reefs on land		
Inland of the coastal zone	7,532	42
Within the coastal zone	2,868	16
Coral rubble on the beach		
Within the coastal zone	5,377	30
Live coral at sea from the reef		
Within the coastal zone	2,282	12
Total	18,059	100

Fishing

Sri Lanka is home to a population of approximately 18 million, with nearly half residing in coastal districts where most economic activities are concentrated. Fishing plays a crucial role in the economy, providing livelihoods for many coastal communities. Fish constitutes around 65% of the animal protein consumed by the population and accounts for 13% of their total protein intake. More than 90% of the fishing population is engaged in the marine sector, which supplied over 97% of the total fish production in the late 1980s . However, it remains unclear how much of this catch is sourced from reef areas. Most fishing activities occur within the first 40 kilometers from the coast .

Historically, fishing methods in Sri Lanka involved non-motorized vessels like dugout canoes and catamarans, employing non-destructive techniques such as angling, gill-netting, and beach seining. In recent years, however, the adoption of highly efficient methods, such as bottom-set nets for catching spiny lobsters and reef fish, has led to significant damage to coral reefs. Additionally, blast fishing, which involves the use of explosives, is prevalent in several regions, particularly in the southern coastal waters around Galle.

Ornamental Fish Collection and Other Exports

Beyond traditional fishing, the collection of reef fish, invertebrates, and live coral for the ornamental fish export industry has gained considerable importance. This sector ranks third in terms of export volume and value, following prawns and lobsters . However, the uncontrolled collection and the use of destructive techniques, such as the 'moxy net' method, can severely damage marine habitats . In addition to ornamental fish, other marine products harvested from coral reef areas include sea cucumbers, spiny lobsters, and various mollusks (e.g., cowries). The collection and export of ornamental fish have surged in recent years, although available data does not provide specific figures on the quantities or species being exported.

Table 04: variation in annual export value (Rs.Million) of ornamental species from Sri Lanka over the period 1962-1994. Source: Wood 1996.

Period	Minimum annual export value in Rupees (million)	Maximum annual export value Rupees (million)
1962 - 1970	0.17	2.16
1971 - 1980	0.09	17.68
1981 - 1990	24.77	96.12
1991 - 1994	93.38	248.30

Tourism and Recreational Activities

Coastal regions in Sri Lanka, especially those with beautiful beaches and fringing reefs, have become key destinations for tourism development. Popular activities include swimming, snorkeling, scuba diving, and viewing coral reefs from glass-bottom boats. However, the rapid growth of tourism infrastructure has often occurred without adequate planning, leading to significant issues related to waste management. In areas like Hikkaduwa, these challenges have intensified, adversely affecting the marine environment.

Environmental Pollution

The disposal of waste into urban waterways and onto beaches poses a serious threat to marine ecosystems, particularly coral reefs. The accumulation of non-biodegradable waste, such as polythene products, in inshore waters has been increasing, especially around cities and coastal towns. Furthermore, untreated industrial effluents are frequently discharged into rivers or canals, which subsequently pollute the sea. Oil pollution in harbors is another chronic issue, exacerbated when fishing vessels are anchored in protected bays formed by fringing reefs or estuaries, where waste oil and bilge water can contaminate the surrounding waters.

Table 05: coral reef locations, status & causes of damage or threats(Major causes of damage are underlined), Source: De Silva, 1985; Rajasuriya & White 1995; Rajasuriya et al.1995;)

Location	Status	Causes of Damage or Threats
Bar Reef Marine Sanctuary	Shallow coral area are partially damaged, deep coral area are in good condition	<u>Crown of Thorns Starfish</u> , destructive Fishing, uncontrolled harvesting, boat anchors
Kandakuliya	Shallow coral areas are heavily damaged	<u>destructive fishing</u> , boat anchors, uncontrolled harvesting
Talawila	Shallow coral areas are damaged	<u>destructive fishing</u> , boat anchors, uncontrolled harvesting
Chilaw	Partially damaged	<u>destructive fishing</u> , <u>uncontrolled harvesting</u>
Negombo	Mostly in good condition, damage is evident, inshore reefs are damaged	<u>destructive fishing</u> , <u>sedimentation on inshore reefs</u> , <u>uncontrolled harvesting</u>
Colombo	Inshore reefs degraded, offshore reefs are in good condition	<u>sedimentation</u> , <u>uncontrolled harvesting</u> , <u>pollution</u>
Ambalangoda to Hikkaduwa	Inshore reefs degraded offshore reef relatively good	<u>coral mining</u> , <u>sedimentation</u> , <u>destructive fishing</u> , <u>uncontrolled harvesting</u>
Hikkaduwa Marine Sanctuary	Partially degraded	<u>sedimentation</u> , <u>boat anchoring</u> , reef trampling, pollution, use of glass bottom boats, increase of Halimeda
Galle including Rumassala reef	Degraded	<u>sedimentation</u> , <u>uncontrolled harvesting</u> , <u>destructive fishing</u> , <u>pollution</u>
Unawatuna	Partially degraded	<u>sedimentation</u> , <u>pollution</u> , reef trampling, anchors, <u>uncontrolled harvesting</u>
Weligama	Partially degraded	<u>sedimentation</u> , <u>pollution</u> , reef trampling, anchors, <u>destructive collecting</u> , <u>uncontrolled harvesting</u>
Polhena, Matara	Heavily degraded	<u>coconut husk seasoning</u> , <u>sedimentation</u> , <u>uncontrolled harvesting</u>
Tangalle	Partially degraded	<u>sedimentation</u> , <u>reef trampling</u> , <u>souvenir collection</u>
Great and Little Basses	Undegraded	<u>uncontrolled harvesting</u>
Batticaloa & Trincomalee	Some locations degraded	<u>destructive fishing</u> , <u>uncontrolled harvesting</u>

Climate change

Coral reefs in Sri Lanka are facing significant threats from various anthropogenic factors, including climate change, pollution, and illegal fishing practices. Dr. Terney Pradeep Kumara, General Manager of the Marine Environment Protection Authority (MEPA), highlighted the dire state of these ecosystems, stating that approximately 90% of the corals in Sri Lankan waters are already dead. He warned that, without urgent intervention, the remaining 10% could be lost within the next decade. This deterioration not only impacts the biodiversity of marine life but also jeopardizes the socio-economic benefits derived from coral reefs, such as tourism and fisheries.

Research indicates that the percentage of live corals in certain regions is alarmingly low. For instance, in Hikkaduwa, the living coral coverage is reported to be around 7%. Additionally, significant coral bleaching events have severely affected the Bar Reef in **Kalpitiya**, while the south's shallow waters predominantly feature coral rubble rather than healthy reefs. Although some coral patches remain in the eastern waters, such as those in Pigeon Islands and the newly designated **Kayankerni** Marine Sanctuary, they are under intense pressure from human activities, including tourism, industrial effluent discharge, agriculture, aquaculture, and urban developments.

Plastic pollution is another critical issue affecting coral health and marine biodiversity. Dr. Pradeep Kumara emphasized the need for societal change regarding plastic consumption, noting that plastic waste significantly harms marine environments. Common items like plastic straws and bottles frequently end up in the ocean, leading to entanglement with corals and contributing to their degradation. Despite existing bans on certain plastic products, such practices continue, exacerbating the challenges faced by coral ecosystems.

07. Conservation efforts

Legislative Protection of Coral Reefs in Sri Lanka

Sri Lanka has established various legislative measures aimed at protecting coral reefs, both directly and indirectly. The two primary laws providing direct protection are the Fisheries and Aquatic Resources Act and the Flora and Fauna Protection Ordinance. However, the fragmentation of protective measures across multiple statutes may lead to overlapping responsibilities and unclear jurisdiction among the relevant authorities.

The Fisheries and Aquatic Resources Act empowers the minister to designate specific marine and land areas as fisheries reserves, thereby prohibiting unauthorized activities such as mining, collecting, and gathering of corals. This is crucial for ensuring that coral growth is not negatively impacted by human activities, particularly for commercial purposes. The Act mandates that individuals obtain permits for coral mining, collection, or processing. While it does allow non-profit organizations and researchers to conduct studies and implement coral rehabilitation efforts, bureaucratic delays can hinder timely and cost-effective restoration initiatives.

The Flora and Fauna Protection Ordinance complements these efforts by permitting the declaration of areas containing coral reefs as sanctuaries, national parks, or nature reserves. Such designations

help control public access and restrict damaging activities, including the removal of corals, which can threaten reef ecosystems.

In addition to these primary laws, coral reefs benefit from indirect protection through other legislation, such as the Coast Conservation and Coastal Resource Management Act. This Act includes provisions for surveying the coastal zone, developing management plans, and regulating activities that could impact coastal ecosystems.

Other Solutions to Protect Coral Reefs in Sri Lanka

Beyond legislative measures, various strategies can be employed to mitigate the destruction of coral reefs. Alternatives to harmful fishing practices, such as using fine mesh barrier nets or line fishing techniques, should be promoted to replace cyanide and dynamite fishing. Implementing stricter regulations for tourism-related activities and addressing pollution from hotels and related businesses can also help reduce coral damage. Additionally, efforts to rehabilitate and restore degraded reefs, such as sinking concrete blocks for support, can facilitate natural regrowth of coral ecosystems.

Education and training programs for communities that rely on coral-related activities, like mining, are essential. The Department of Coast Conservation and Coastal Resource Management has initiated programs to educate and train both law enforcement and local communities in coastal areas.

Table 06: Management status of Sri Lanka Marine Protected Areas.

Name	Year of declaration	Gazette no	Extent (ha)	Remarks	Management plan
Adam's Bridge Marine National Park	2015	1920/3 of 22.06.2015	18,900	100% marine area	No
Pigeon Island Marine National Park	2003	1291/16 of 04.06.2003	471.43	100% marine	Yes
Hikkaduwa Marine National Park	2002	1257/14 of 08.10.2002	250	100% marine	Yes (Special Area Management Plan under SAMP)
Delft Island National Park	2015	1920/3 of 22.06.2015	1,846	100 m wide belt in the sea. Rest is land	No
Chundikulam National Park	2015	1920/3 of 22.06.2015	19,565.33	43% marine area in coastal waters	No
Vidattalivu Nature Reserve	2016	1956/13 of 01.03. 2016	29,180	77% marine area	No
Nagarkovil Nature Reserve	2016	1956/13 of 01.03. 2016	7882	66% marine area in coastal waters	No
Nayaru Nature Reserve	2017	2003/10 of 24.01.2017	4464.35	25% lagoon. (This PA has a narrow strip of the coastal waters)	No
Bar Reef Sanctuary	1992	708/24 of 03.04.1992	30,670	100% marine	Yes (Special Area Management Plan under SAMP)
Vankalai Sanctuary	2008	No. 1566/3 of 08.09.2008	4838.95	62% marine	No
Paratitivu Paratitivu Island Sanctuary	1973	No. 60 of 18.05.1973	97.1	100% marine	No
Kayankerni Sanctuary	2019	2118/59 of 11.04.2019	953.25	99% marine	No
Great and Little Sober Islands Sanctuary	1963	21.06.1963	64.7	100% marine	No
Rekawa Sanctuary	2006	25.05.2006	271	83% marine (includes 500 m belt of sea)	No
Godawaya Sanctuary	2006	25.05.2006	230.99	83% marine (includes 500 m wide belt of sea)	No
Kalametiya Sanctuary	1984	28.06.1984	2,525.2	13% marine	No
Great and Little Busses Sanctuary	2019	2144/60 of 11.10.2019	67,282.3	100% marine	No
Rumassala Sanctuary	2003	03.01.2003	170.7	90% marine	No

08. Acts and regulations in Sri Lanka

1. Coast Conservation and Coastal Resource Management Act, No. 57 of 1981 (Amended in 1988 & 2011)

- ✓ Creates the Coast Conservation and Coastal Resource Management Department (CCCRMD).
- ✓ Offers a legal mandate for drafting Coastal Zone Management Plans (CZMP).
- ✓ Able to issue permits for all construction or development within the coastal zone

2. National Environmental Act, No. 47 of 1980 (Amended in 1988 & 2000)

- ✓ Implemented by the Central Environmental Authority (CEA).
- ✓ Controls on polluting, waste disposal, EIA.
- ✓ Safeguards marine ecosystems, mangroves, and wetlands.

3. Marine Pollution Prevention Act, No. 35 of 2008

- ✓ It is governed by Marine Environment Protection Authority (MEPA).
- ✓ It bans the discharge of pollutants (oil, chemicals, waste) into the sea.
- ✓ Regulate ship-based pollution and oil spill responses.

4. Fisheries and Aquatic Resources Act, No. 2 of 1996

- ✓ Administered by the Department of Fisheries and Aquatic Resources
- ✓ Regulates fishing activity, the mining of corals and destruction of marine habitats.
- ✓ From October 2023, it is mandatory to apply for permits if you want to fish in certain protected areas.

5. Fauna and Flora Protection Ordinance, No. 2 of 1937 (Amended in 2009)

- ✓ Conserves marine species and coastal ecosystems.
- ✓ Overseeing activities in marine national parks and sanctuaries

6. Forest Ordinance, No. 16 of 1907 (Amended in 2009)

- ✓ It protects mangrove and coastal forests.
- ✓ Regulates deforestation and land-use changes near coastal areas in Sri Lanka.

7. Urban Development Authority (UDA) Law, No. 41 of 1978

- ✓ Mention the Coastal Commission, which has to approve coastal development projects.

3. Acts on other countries

1. Laws related to marine spatial planning (MSP)

- ✓ MSP Laws Per Country: United States, Australia, and European countries
- ✓ National Marine Spatial Planning (MSP) laws are existing in many countries that support the sustainable use of coastal and marine resources.
- ✓ There is no legally backed MSP regime in Sri Lanka. Instead, various agencies manage coastal areas in silos, creating conflicts and inefficiencies.

2.Increased Mangrove Protection Laws

- ✓ Countries with rigid laws: Indonesia, Philippines, Thailand
- ✓ Take Indonesia, where strict laws ban the clearing of mangroves with large penalties for violators.
- ✓ Sri Lanka has mangrove conservation policies and protects many mangroves under its forest laws, but it lacks a dedicated Mangrove Protection Act and enforcement is weak.

3.Plastic Ban Strict Laws Bombarding Coastal Areas

- ✓ Countries with strict restrictions: Canada, Kenya, European Union nations
- ✓ Some countries have strict bans on plastic, particularly in coastal and marine environments.
- ✓ Sri Lanka has single-use plastic regulations, but enforcement is weak and microplastic pollution in coastal areas remains a major problem.

4.Blue Carbon Legislation

- ✓ Laws concerning Blue Carbon by Country: Australia, USA
- ✓ Multiple countries have also recognised and regulated blue carbon ecosystems (mangroves, seagrass beds) in the context for climate change mitigation.
- ✓ There are no specific policies guiding the development of blue carbon projects in Sri Lanka.

6.Stronger Coastal Setback Laws

- ✓ Countries with strict setback laws: India, USA, Maldives
- ✓ In countries like India, there are stringent Coastal Regulation Zones (CRZ) to prevent construction too close to the coast.
- ✓ There are setback rules in Sri Lanka, but enforcement is lax, and illegal coastal construction is frequent.

7.Coral Reefs: Laws and Mechanisms for Their Protection

- ✓ Countries with strict setback laws – Australia (Great Barrier Reef Act), USA (Coral Reef Conservation Act)
- ✓ Coral protection in Sri Lanka falls under the general environmental laws and there is no separate Coral Protection Act like is in Australia or the USA.

8.Stricter Oil Spill and Ship Waste Regulations

- ✓ Decent marine pollution laws: USA (OPA 1990), EU (MARPOL Enforcement)
 - The Oil Pollution Act (OPA 1990), USA, provides stringent penalties for oil spills.
 - MARPOL compliance is in place but enforcement of such compliance and the emergency response mechanisms are weak in Sri Lanka.
- ✓ Participatory Governance in Coastal Management
- ✓ Countries with high public engagement: Norway, Canada
 - In some countries, citizens are allowed to participate in decisions about coastal development, Coastal area management in Sri Lanka is government centric with little participation by communities.

METHODOLOGY

This study focuses on assessing coral and coastal pollution in selected regions, emphasizing the impact of pollution on coral reefs and mangrove ecosystems. The methodology includes field visits, data collection, visual assessments, expert consultations, and community feedback analysis.

1. Field Visits and Site Selection

Our team visited multiple locations to study pollution impacts:

- Salli Amman Kovil Beach Side
- Kevuliya Beach Side
- Sampathivu Mangrove Areas
- Inner Harbour road beachside
- Thirukadalur beachside

During these visits, we documented environmental conditions, collected samples, and engaged in visual assessments.

2. Visual Analysis

- **Coral Reefs:** Dead corals turned white post-mortem, while some exhibited a brownish.
- **Mangrove Plants:** Healthy and damaged mangrove areas were identified in Sampathivu.
- **Coastal Pollution:** Plastic, Glass waste, polythene, biodegradable waste (dead fish, dead algae, coconut shells), and construction waste were observed and recorded.
- **Water Quality:** Visually identified contamination in the waters at Sampathivu
- **Coastal Erosion:** Observed at Salli Amman Kovil Beach, Inner Harbour road beachside, Thirukadalur beach side where black stone barricades were placed for protection.
- **Beach cleanup programs:** As part of our data collection, we documented the Clean Sri Lanka beach cleanup program through photographs. The program successfully collected a significant amount of waste, which was used to assess the extent of pollution in the coastal area.

3. Expert Consultations

- **Coast Conservation and Coastal Resource Management Department (Trincomalee)**
 - Insights on coral damage documented from a project (2010-2016).
 - Discussion on the Pigeon Island Management Plan and zoning regulations.
 - Details on artificial coral planting initiatives.
 - Clean Sri Lanka Project and its future coastal clean-up extensions.
 - Information on erosion patterns and seasonal effects in Trincomalee.
 - Plastic waste management initiatives.

- Impact of building restrictions near coastal regions.
- Factors affecting beach design and shape variations.
- Awareness programs on coastal conservation.
- Collected proof letters on corrosion complaints.
- Agricultural runoff mixing in the irrakkandy side.
- **Marine Environmental Protection Authority (MEPA)**
 - Data on water quality parameters such as pH, salinity, TDS (Total Dissolved Oxygen), Temperature, conductivity, turbidity and resistance.
 - Discussions on ocean acidification and global warming as a primary cause of coral discoloration.
 - Review of artificial coral planting projects.
 - Analysis of sunscreen chemicals affecting coral health.
 - Observations on beach clean-up programs under the Clean Sri Lanka Project.
 - Identification of highly polluted areas, particularly Deer Park Beach.
 - Provided details on areas where the Clean Sri Lanka Project conducted beach clean-ups in Trincomalee.

4. Community Feedback and Awareness

A survey was conducted using Google Forms targeting individuals who frequently visit beaches in Trincomalee. The survey aimed to understand public perception of pollution, its effects on marine life, and awareness of conservation efforts.

TIMELINE

Week	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th	13th	14th
Discuss the topic and objectives														
Analyzing topic, set goals & objectives														
Preparing the proposal														
Identifying the data resources														
Planned for field visit														
Analyzing data's														
Preparing mid-semester presentation, report														
Collecting data by field visit 1														
Investigation & data analysis														
Collecting data by field visit 2														
Preparing final report & presentation														

FIELD OBSERVATION AND FINDINGS

1. Coral Reefs

- Dead coral samples were predominantly white, indicating bleaching, while some showed brownish discoloration.
- Artificial coral planting was identified as an initiative to restore reef health.
- Rising sea temperatures were recognized as a major contributor to coral bleaching.

2. Mangrove Ecosystem

- Healthy and damaged mangroves were recorded in Sampathivu.
- Contaminated water and dead aquatic life were observed in selected areas.
- Agricultural waste, such as castor oil residues, was detected in coastal regions.
- Soil texture variations were noted as a factor affecting coastal development and erosion resistance.

3. Coastal Pollution and Erosion

- Large amounts of plastic and polythene waste were found at Kevuliya and Salli Amman Kovil Beach.
- Seasonal erosion in Trincomalee was identified, particularly in Thirukadalur, Nilaveli, Muttur, Inner Harbour beachside, Salli Aman Kovil near beachside and Kinniya.
- Black stone barricades were used in erosion-sensitive areas to mitigate coastal degradation.
- The impact of building construction waste was visible near Salli Amman Kovil Beach.
- Details of the areas where the Clean Sri Lanka Project conducted beach clean-ups in Trincomalee were collected.
- Photos of Clean Sri Lanka Project beach cleanup activities were taken to document the beach clean-ups in Trincomalee.

4. Water and Air Quality

- Water samples showed contamination with pollutants.
- pH, salinity, TDS (Total Dissolved Oxygen), Temperature, conductivity, turbidity and resistance data were collected from MEPA for further analysis.
- Air quality was monitored using the Windy App, with no significant pollution detected.

5. Conservation Initiatives and Challenges

- Limited enforcement of zoning laws for coastal protection.
- Need for increased awareness on plastic waste management and sustainable tourism practices.

- Potential benefits of replanting mangroves to prevent erosion.
- Importance of continued beach clean-ups under the Clean Sri Lanka Project.
- Monitoring of seasonal changes and implementing long-term conservation strategies.

DATA ANALYSIS AND INTERPRETATION

1. Plastic and Biodegradable Waste Analysis

During field visits to **Salli Amman Kovil, Kevuliya, and Sampathivu beach sides**, different types of plastic and biodegradable waste were observed. The collected data provides an overview of the waste pollution affecting these coastal areas.

1. Breakdown of Observed Waste

The waste materials identified were categorized into two main types:

A. Plastic Waste

- Plastic bottles – Present at all three locations.
- Polythene (Regiform and other plastic waste) – Found at Salli Amman Kovil and Sampathivu.
- Plastic cups – Observed in Sampathivu.
- Unused slippers – Noted in Kevuliya and Salli Amman Kovil.
- Glass bottles – Found in Kevuliya and Sampathivu.

B. Biodegradable Waste

- Coconut tree-related waste – Found in Kevuliya and Salli Amman Kovil.
- Dead coral – Observed in large amounts at Kevuliya and Salli Amman Kovil.
- Algae – Present in Kevuliya.
- Dead fish – Observed at Salli Amman Kovil.
- Dead oyster shells – Found in large amounts at Sampathivu.

2. Site-Wise Waste Distribution

Beach Site	Plastic Waste Observed	Biodegradable Waste Observed	Other Observations
Kevuliya	Plastic bottles, unused slippers, glass bottles	Coconut tree-related waste, dead coral (visually large), algae	-
Salli Amman Kovil	Plastic bottles, polythene, unused slippers	Coconut tree-related waste, dead coral (visually large), dead fish	Construction waste

Beach Site	Plastic Waste Observed	Biodegradable Waste Observed	Other Observations
Sampathivu (Mangrove side)	Plastic bottles, polythene, plastic cups, glass bottles	Dead oyster shells (visually large)	Contaminated water (visibly unclear)

3. Key Observations

- Plastic bottles were present at all three locations, indicating widespread plastic waste pollution.
- Dead coral was notably high in Kevuliya and Salli Amman Kovil, suggesting environmental stress in these regions.
- Sampathivu exhibited visibly contaminated water, possibly due to pollution from surrounding activities.
- Biodegradable waste such as coconut husks and dead fish was more prominent in Kevuliya and Salli Amman Kovil, potentially influenced by natural factors and human activity.



Figure 05: visual analysis of field visit sampathivu



Figure 05: visual analysis of field visit salli amman kovil beach side

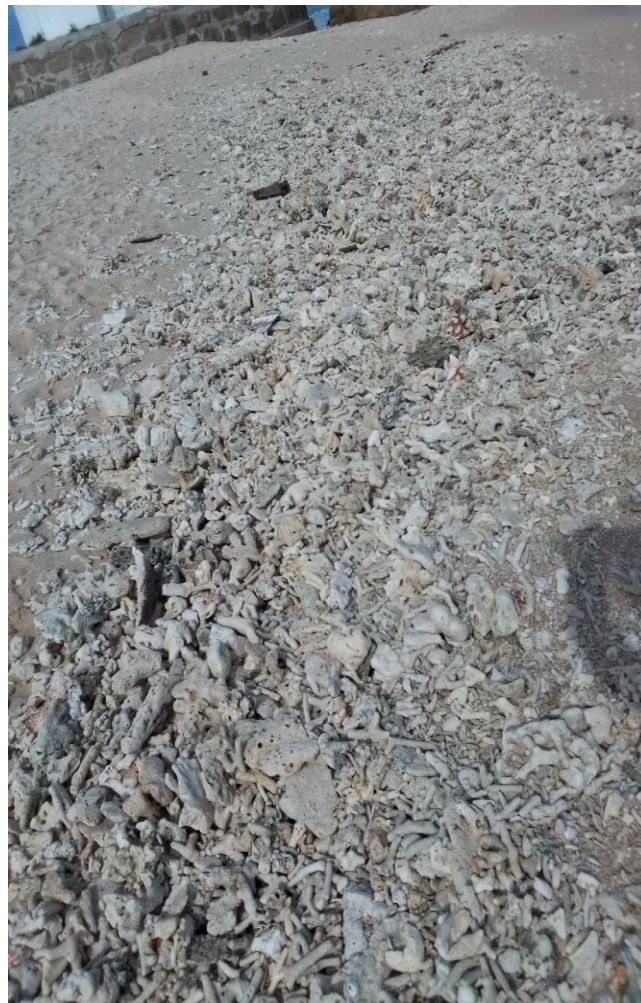


Figure 05: visual analysis of field visit I beach side

2.Coral Health (Coral Discoloration)

We looked at the coral reefs health during our visits to Salli Amman Kovil and Kevuliya, paying special attention to how the color changed in corals that had died. We mainly saw two things: coral bleaching and coral browning.

Coral Bleaching: We noticed a lot of dead corals that had turned white. This whitening, or bleaching, happens when corals get stressed out and kick out the tiny algae (called zooxanthellae) that live in them. These algae give corals their color and are super important for their survival. Things like warmer ocean temperatures, pollution, or other environmental problems can stress the corals out. When the algae leave, you can see the coral's white skeleton underneath, which means the coral is dead.

Coral Browning: Some of the dead corals had turned a brownish color. This browning usually happens when dead corals start to collect things like organic matter, tiny living organisms, or sediment on their skeletons. Over time, this buildup makes the skeleton look brown. Browning can also be a sign that the coral is breaking down after it dies.

Interpretation:

- The bleaching observed in dead corals indicates extreme environmental stress, likely from factors such as elevated water temperatures or pollution. Since bleached corals are often no longer alive, the ecosystem in these areas is under significant threat.
- The browning of dead corals suggests that, over time, there may be continued degradation of the coral skeletons due to the accumulation of organic material or microbial growth.
- The combination of bleached and browned corals in these areas highlights the urgency of addressing environmental stressors to prevent further coral loss. It underscores the need for further research and effective conservation measures to protect coral ecosystems.





Figure 06: Death coral visual analysis of Salli Amman Kovil and Kevuliya beach side

3.Mangrove Health

In the Sampathivu area, experts took a close look at how healthy the mangrove plants were, paying special attention to their leaves. Here's what we found:

Leaves in Good Shape: Some mangrove plants had leaves that were growing normally and looked healthy. This is a good sign, suggesting these plants are doing well and getting all the nutrients and right conditions they need to thrive.

Leaves Not Growing Properly: Other mangrove plants had leaves that weren't growing as they should. This probably means the plants are under some kind of stress or aren't happy with their surroundings. These leaves might be stunted because the plants aren't getting enough nutrients, are struggling with too much salt in the water, or are being disturbed by something in the environment.

Interpretation:

- The presence of fully growable leaves indicates that certain mangrove plants in Sampathivu are adapting well to their environment, maintaining healthy growth and strong health.
- The observation of non-growable leaves suggests that some mangrove plants are experiencing stress, likely due to environmental factors such as changes in water salinity, pollution, or nutrient deficiencies.
- These observations highlight the overall health of the mangrove ecosystem in Sampathivu, suggesting that while some areas are thriving, others may be under stress, which could impact the long-term health and stability of the mangrove forest



Figure 07: Sampathivu mangrove plants visual analysis

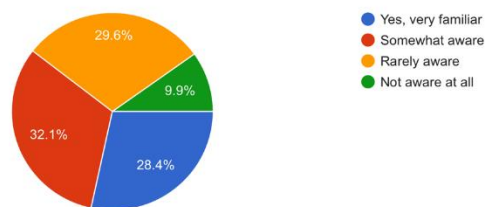
4. Community Feedback

The survey carried out within the Trincomalee community offered valuable insights into public awareness, perceptions, and concerns about coral and coastal pollution. The responses shed light on major environmental challenges and potential solutions for improved conservation efforts.

1. Community Awareness and Perception

- Most people surveyed knew that Trincomalee is a main spot for coral reefs, showing they have a decent understanding of marine life diversity.
- There's still more to learn, as some weren't sure where corals grow, suggesting we need to focus on teaching specific information.

6. Are you aware of any local efforts to protect Trincomalee's coral reefs?
81 responses



Graph:01

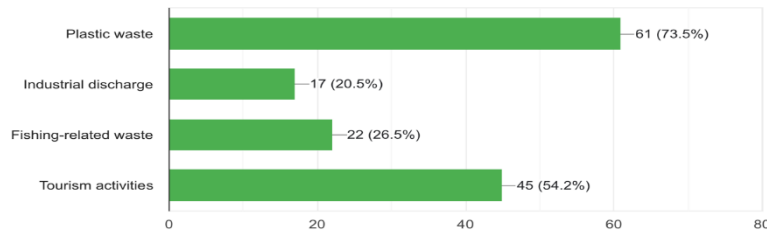
2. Pollution Concerns and Sources

- Plastic waste emerged as the dominant pollution concern, with fewer concerns about industrial discharge and fishing-related waste.

- Over 63% of respondents observed coral bleaching occasionally or frequently, highlighting how environmental pressures create noticeable changes.
- 50% rated the government's efforts as either poor or extremely poor, which indicates

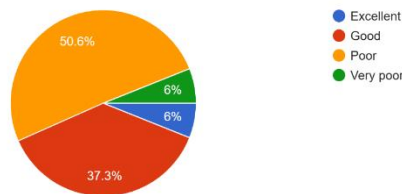
3. What do you think is the main source of pollution in Trincomalee's coastal areas?

83 responses



13. How do you rate the government's role in addressing coastal pollution in Trincomalee?

83 responses



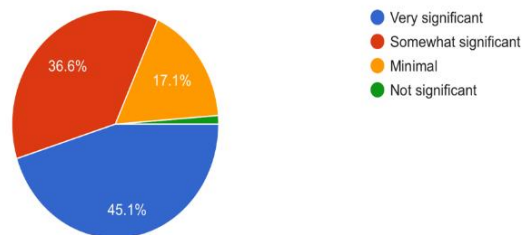
Graphs :02

3. Role of Tourism in Pollution

- Almost half of the people surveyed think that tourism has a big effect on pollution, and 36% see it as having a somewhat noticeable impact.
- The general public tends to support eco-tourism initiatives as a way to reduce damage to the environment.

5. How significant do you think the impact of tourism is on coastal pollution?

82 responses



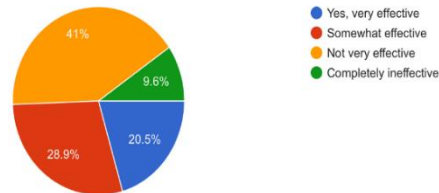
Graph :03

4. Community Participation in Conservation

- While 40% have participated in cleanup programs at least once, regular engagement remains low.

- Around 60% of those surveyed say they're just kind of aware of local conservation efforts, and only 28% feel they know a lot about them. This suggests we need to do a better job of getting the word out.

9. Do you think plastic waste management in Trincomalee is effective?
83 responses

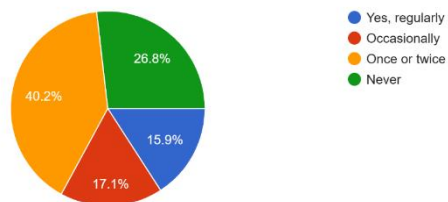


Graph:04

5. Waste Management and Regulations

- Most people think that waste disposal facilities aren't good enough, which really shows that we need better infrastructure right away.
- 70% percent of those surveyed are in favor of tighter rules, indicating that the public largely supports new laws aimed at safeguarding coastal environments.

10. Have you participated in any beach cleanup initiatives in Trincomalee?
82 responses



Graph :05

6. Corrosion in Trincomalee

1. Visible signs

To understand erosion in Trincomalee, we visited the following locations:

- Inner Harbour Road beachside
- Thirukadalur beachside
- Salli Amman Kovil beachside

During our visit, we took photos of these areas and observed the protective measures in place to prevent coastal erosion.

Location	Protective Measures	Condition of Barricades	Observations
Inner Harbour Road Beach	Black stones, large stone barricades	Some barricades damaged	Signs of erosion visible
Thirukadalur Beach	Large stone barricades	It's mostly unchanged.	Moderate erosion observed
Salli Amman Kovil Beach	Black stones, barricades	Some damaged barricades	Protective measures in place, but erosion still present

2. Corrosion complaints

To further support our case, we've gathered letters of complaint regarding corrosion, which demonstrate that Trincomalee is indeed dealing with this issue. These letters act as solid evidence, reinforcing what we've noticed about the coastal erosion happening there and highlighting the importance of taking action to protect the area.

Date	Source	Location	Complaints Summary
April 2023	Coastal Conservation and Coastal Management Department	Ralkuli, Thaqwa Nagar, Nadutheeva, Kadtakaraichenai	Erosion has gradually advanced, consuming agricultural and domestic land, and widening over time.
15 July 2023	Local Authority of Muthur Habbip Area	Muthur Habbip Area	Coastal erosion reported, with protective measures implemented. The effectiveness of measures was not detailed in the letter.

பிரதேச செயலகம் - மூதூர்
 Divisional Secretariat - Muthur
 026 - 2238235
 026 - 2238237
 026 - 2238668
 026 - 2238668
 026 - 2238667

மூதூர் பிரதேச செயலகம்
 Muthur Divisional Secretariat
 2023.03.15
 பிரதேச செயலகம் - மூதூர்
 Divisional Secretariat - Muthur
 E-mail: ps.muthur@gmail.com
 Web site: www.muthur.sarha.gov.lk

Area Engineer
 Coast Conservation & Coastal Management Department
 Nillavelli
 Trincomalee.

Protection of Coastal Erosion

Referring to the District Coordinating Committee meeting held on March 28th, 2023 at the District Secretariat in Trincomalee regarding the protection of coastal erosion, it was noted that the Ralkuli, Thaqua Nagar, Naduthevu, and Kadatkarachenai coastal Grama Niladhari divisions are experiencing a slow but steady advance of erosion. Over the years, the width of erosion has gradually increased, and the sea has already consumed vast areas of agricultural and domestic land in the aforementioned coastal Grama Niladhari divisions.

Therefore, appropriate action needs to be taken to address this pressing matter.

M.B.M.Mubarak
 Divisional Secretary
 Muthur

Copy : Government Agent / District Secretary, District Secretariat, Trincomalee.

மூதூர் பிரதேச சபை
 MUTUR PRADESIYA SARHA
 இவர் பூர்த்தியை விரைவில்
 20.07.2023
 094-0262238374
 094-0262238376
 Reception: +94-026-2238573
 E-mail: psmuthur@yahoo.com

My No: MUPS/ Secretary/Gov.Reg23 Your No: Date: 2023.07.15

உள்நாட்டி ஆணையாளர்,
 உள்நாட்டி ஆணையாளர் அலுவலகம்,
 மிகக்கு மாகாணம்.

கொடிய அழகுடன் கூடிய கட்டிடங்களைக் கட்டி மூதூர் ஸ்டேட் சபைகளில் உடனடி
 ஏற்பாடுகள் பராமுகமாக மேற்கொள்ளப்படும்.

மேற்படி தங்களுக்கு EP/14/D6/Gen/2023, இலக்க 2023.07.05 ஆம் திகதிக்கு கடிதம் தொடர்பாக,
 இவ்வெண்ணெய்திட்டத்தினை மேற்கொள்வதற்கு எம்மிடம் நிதி வசதி இல்லாத காரணத்தினால், கனரகப்போர்
 பாதகமாகத் திணைக்களம் பொறிப்பாளரைத் தொடர்பு கொண்டு இவ்வெண்ணெய்திட்டத்தை மேற்கொள்ள
 முடியுமாயின் மேற் கொண்டு தகுமாறு நடவடிக்கை மேற்கொள்ளப்படும்.

2023.07.07 ஆம் திகதி அமைந்த மூதூர் ஸ்டேட் சபைத் தங்களுக்கு இடத்தினைப் பற்றியும், குறித்த
 இடத்தில் தற்போதைய கால நிலையில் உடனடி நடவடிக்கை மேற்கொள்ளப்படும். இவ்வெண்ணெய்திட்டத்தை
 மேற்கொள்ளும் இவ்வெண்ணெய்திட்டத்திற்கு நிதி ஒதுக்கீடு மேற் கொள்ளப்படாத காரணத்தினால், கனரகப்போர்
 நிலைமைகளில் பாதகமாகத் திணைக்களம் பொறிப்பாளரைத் தொடர்பு கொண்டு இவ்வெண்ணெய்திட்டத்தை மேற்கொள்ள
 முடியுமாயின் மேற் கொண்டு தகுமாறு நடவடிக்கை மேற்கொள்ளப்படும்.

Y. GATEYASOTTI
 Secretary
 Muthur,
 Pradeshiya Sabha

Figure 03: Coastal erosion complaint letters

3. How Coastal Areas Are Assessed for Vulnerability and Setback Measurement

- **Identifying Coastal Segments:** The coastline is divided into segments based on location and environmental conditions.
- **Assessing Vulnerability:** Factors like erosion risk, wave action, and environmental sensitivity determine if an area is **low, medium, or high vulnerability**.
- **Measuring Setback Distances:**
 - **Reservation Area (No Build Zone)** – Closest to the sea, highly restricted.
 - **Restricted Area (Soft Zone)** – Allows limited development.
 - **Total Setback** – Sum of both areas, ensuring safety and conservation.
- **Applying Regulations:** Higher vulnerability requires **larger setbacks** to protect against environmental damage.

	Level of Vulnerability	Coastal Segments Nos. 1- 44			Coastal Segment Nos. 45-105		
		Reservation Area	Restricted Area	Total Setback	Reservation Area	Restricted Area	Total Setback
	Low (-)	10	25	35	20	30	50
	Low (+)	15	25	40	25	40	65
	Medium (-)	15	30	45	30	50	80
	Medium (+)	20	30	50	35	60	95
	High (-)	20	35	55	40	70	110
	High (+)	25	35	60	45	80	125
		Protected Areas			Protected Areas		
		Conservation Zones			Conservation Zones		

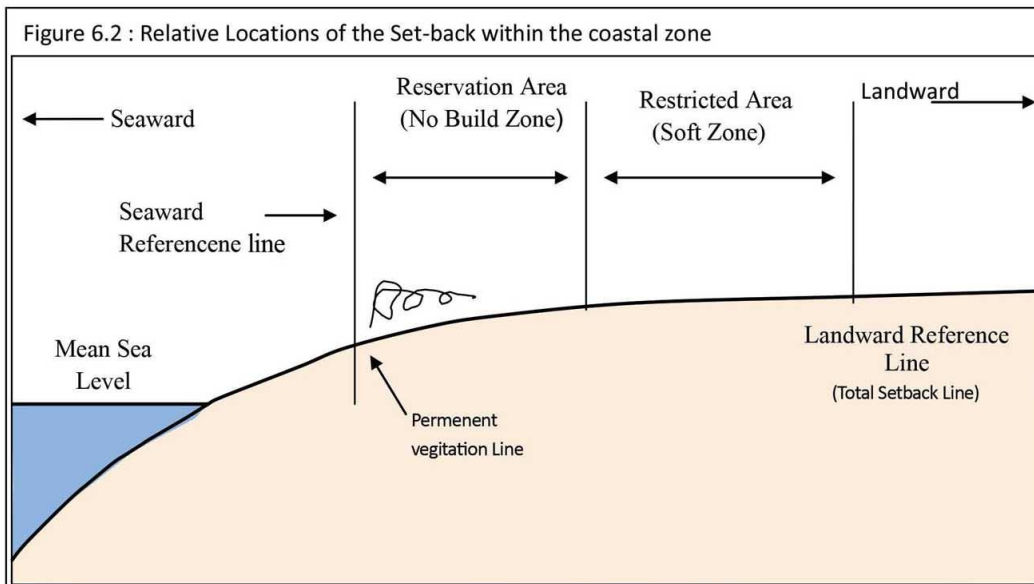


Figure 04: Vulnerability and Setback Measurement

4. Trincomalee Coastal Areas and Their Setback Distances

Segment	Vulnerability Level	Reservation Area (m)	Restricted Area (m)	Total Setback (m)
Lankapatuna to Foul Point	Medium (+)	35	60	95
Foul Point to Muthur East River Mouth	Low (-)	20	30	50
Muthur East River Mouth to Gangei Bridge	Medium (-)	30	50	80
Gangei Bridge to Irrakkandy Bridge	Low (-)	20	30	50
Irrakkandy bridge (River Mouth) to Slapayaru Bridge	Medium (-)	25	40	65
Kuchchaveli Puduwakattu to Kallarawa (close to Thiriyaya Junction)	Low (-)	30	50	80
Kuchchaveli Puduwakattu to Kallarawa (close to Thiriyaya Junction)	Low (+)	25	40	65
Kallarawa (Close to Thiriyaya Junction) to Pulmude Arisimale Point	Medium (-)	30	50	80
Pulmade Arisiamale point to Kokilai Lagoon Southern Boundary	High (+)	45	80	125

5. Project Completed Proof for Coastal Erosion Protection

Coastal erosion poses a significant threat to infrastructure and communities in Trincomalee. Various projects have been undertaken to mitigate its effects and protect key locations from damage. The following are two major completed projects aimed at addressing this issue:

1. **Protection of Muthur Fishing Village and Naval Base from Coastal Erosion**
 - Coastal erosion has been a persistent issue affecting the Muthur fishing village and the naval base. The implemented project involved constructing protective barriers to safeguard these areas from further damage. The photos attached provide visual evidence of the completed protective structures.
2. **High Tension Line and CEB Tower Protection Project at Upparu, Kinniya**
 - A high-voltage transmission tower was initially constructed on land near the Upparu Bridge. However, due to ongoing coastal erosion, the tower is now located in the middle of the sea. If this tower sustains damage, the electricity supply to thousands of homes and buildings in Muthur and Kinniya will be severely impacted.
 - To prevent potential structural failure, our department has implemented security structure construction. The attached photographs document the ongoing and completed measures taken to reinforce the area and ensure the stability of the tower.

These projects demonstrate proactive efforts to fight coastal erosion and protect critical infrastructure in Trincomalee.

Construction of 100m Gabion Wall - Muthur (Trincomalee)		
Purpose / Objectives: Protection of Muthur fishing village and naval base from coastal erosion		
Location: Muthur		District: Trincomalee
Allocation Source: GOSL Allocations		
Date of Commencement: 2024-05-01	Date of Completion: 2024-08-31	Project Duration: 04 Months
Total cost of the Project: Rs.Mn.10.0		Project Cost for 2024: Rs. Mn 10.0
Beneficiaries: Local community, fishing community and infrastructure		
Present Situation 		

Figure 05: project plan in Muthur coastal area



Figure 06: project plan in Upparu, Kinniya coastal area

7. Overview of Water Quality Parameters

Nilevali Beach – 2024

Month	Site	Temperature (°C)	pH	Conductivity (µS/cm)	TDS (mg/L)	Salinity (ppt)	Turbidity (NTU)	Resistance (Ohm/cm)
Jan	1	29	5.95	51.9	26.06	33.61	11	18.66
Mar	2	29	6.04	53.19	26.2	34.68	10	18.8
	3	29	5.96	53.24	26.18	33.96	10	18.7
	4	29	6.09	53.48	26.19	33.98	9	18.69
	1	28.01	6.29	55.08	26.99	36.72	6	18.16
	2	27.6	6.27	54.96	26.96	36.67	5	18.15
April	3	27	6.3	55.1	26.88	36.76	5	18.19
	4	27.3	6.3	55.12	26.79	36.7	4	18.16
	1	31.3	6.7	52.08	25.52	34.54	7	19.2
	2	31.1	6.71	52.3	25.64	34.77	9	19.11
	3	31.2	6.44	52.4	25.67	34.77	8	19.09
May	4	31.4	6.69	52.3	25.65	34.7	6	19.1
	1	31.4	6.21	50.92	25.06	33.85	3	19.55
	2	31.4	6.24	50.91	25.08	33.9	4	19.56
June	3	31	6.27	50.9	25.09	33.95	6	19.6
	4	32	6.28	50.96	25.09	33.97	7	19.61
	1	29.1	6.23	5094	25.07	33.85	6	19.47

Month	Site	Temperature (°C)	pH	Conductivity (μS/cm)	TDS (mg/L)	Salinity (ppt)	Turbidity (NTU)	Resistance (Ohm/cm)
July	2	29	6.18	50.82	25.09	33.73	4	19.43
	3	29.2	6.16	50.91	25.17	33.64	3	19.47
	4	29.6	6.29	50.96	25.42	33.81	3	19.39
	1	28.7	6.07	51.4	25.19	33.98	5	19.44
	2	28.6	6.02	51.38	25.16	33.96	8	19.41
	3	28.9	6.06	51.42	25.17	33.91	4	19.38
	4	29	6.01	51.37	25.3	33.89	3	19.37
Aug	1	30	7.41	49.49	21.22	32.57	16	20.24
	2	30.2	7.39	50.12	24.56	33.05	20	19.95
	3	30.3	7.4	50.14	24.57	33.08	26	19.94
Oct	4	30.7	7.31	50.18	24.52	33.18	22	19.68
	1	26.8	7.0	64.46	32.05	42.7	-	15.48
	2	27.1	6.98	64.11	32.16	40.1	-	15.42
	3	26.9	6.63	63.27	32.17	42.63	-	15.36

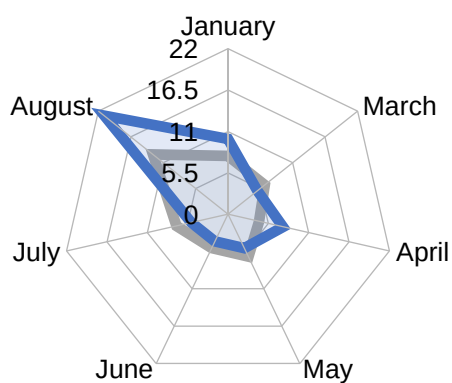
Trincomalee Beach – 2024

Month	Site	Temperature (°C)	pH	Conductivity (µS/cm)	TDS (mg/L)	Salinity (ppt)	Turbidity (NTU)	Turbidity (Ohm/cm)
Jan	1	29.0	5.77	48.8	23.91	30.18	6	20.49
	2	29.1	5.79	49.01	23.88	30.12	9	20.5
	3	29.0	5.56	51.58	25.27	30.54	11	19.39
	4	29.0	5.80	51.1	25.49	30.36	5	19.1
Mar	1	29.0	6.25	51.1	25.29	34.26	5	18.39
	2	29.0	6.27	54.38	26.54	32.95	4	18.43
	3	29.0	6.30	54.46	26.66	36.25	9	18.43
Apr	4	30.8	6.30	54.66	26.6	39.5	7	18.29
	1	30.6	6.30	52.38	22.6	36.16	4	18.23
	2	30.5	6.31	50.13	23.97	32.18	4	22.14
	3	30.6	6.30	48.87	23.49	32.26	3	24.46
	4	32.7	6.35	49.99	25.97	31.39	6	20.44
May	1	32.5	6.29	51.15	25.06	33.85	5	23.01
	2	31.2	6.36	51.1	25.05	33.8	7	19.55
	3	31.4	6.35	51.3	25.08	33.48	6	19.48
	4	31.4	6.35	51.4	25.07	33.22	7	19.47
June	1	29.1	6.29	51.04	25.01	33.72	4	19.61
	2	29.3	6.21	51.21	25.02	33.64	5	19.11
	3	29.2	6.10	51.33	25.12	33.21	5	19.27

Month	Site	Temperature (°C)	pH	Conductivity (µS/cm)	TDS (mg/L)	Salinity (ppt)	Turbidity (NTU)	Turbidity (Ohm/cm)
	4	29.4	6.26	50.63	24.8	33.61	6	19.63
July	1	29.1	6.26	50.6	24.89	33.39	3	19.76
	2	29.0	6.14	50.48	24.76	33.64	10	19.71
	3	29.1	6.25	49.96	24.49	33.37	7	19.73
	4	29.27	6.18	49.95	24.48	33.32	8	19.75
Aug	1	29.1	7.27	49.99	24.46	32.92	16	20.01
	2	29.27	7.30	49.98	24.45	32.93	17	20.02
	3	29.2	7.32	49.96	24.44	32.91	10	20.02
	4	29.3	7.28	49.95	24.46	32.16	8	20.02
Oct	1	25.6	8.95	67.61	33.03	44.27	-	14.72
	2	25.6	6.84	66.33	32.09	44.05	-	14.32
	3	26.2	6.92	65.24	33.13	44.21	-	14.71
	4	24.9	6.73	67.43	33.18	44.26	-	14.26

TURBIDITY

- Turbidity (NTU) Trincomalee beach
- Turbidity (NTU) Nilaveli beach

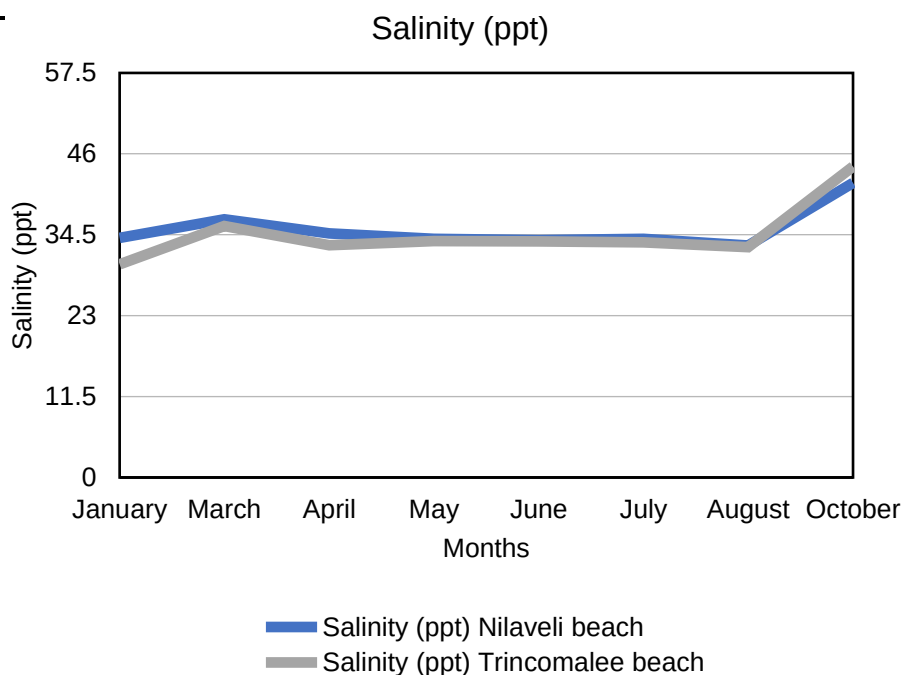


Graph:06

Turbidity in sewage refers to the cloudiness of waste water made by unwanted solids and microorganism it is an important parameter to check the water quality. it is reducing the light going through the water.

Average turbidity of site nilaveli beach is 8.21NTU, Average turbidity of Trincomalee beach side is 7.03NTU according to this data averagely this is secondary rated sewage. these sewages are allowable to mixed with ocean. By comparing turbidity data's of Nilaveli beach and Trincomalee beach ,In March, May, June, July Nilaveli's turbidity is lower than Trincomalee's turbidity But in January and August it is not like that . specifically in August both places are looking for higher turbidity. Nilavei got 21NTU and Trincomalee got 12.75 And the difference is 8.75. Usually, higher amount of tourists visit Trincomalee in August month and nilaveli is the most attractive place for tourists. so according to this data's, we assume that Turbidity in sewage is increase drastically by tourism.

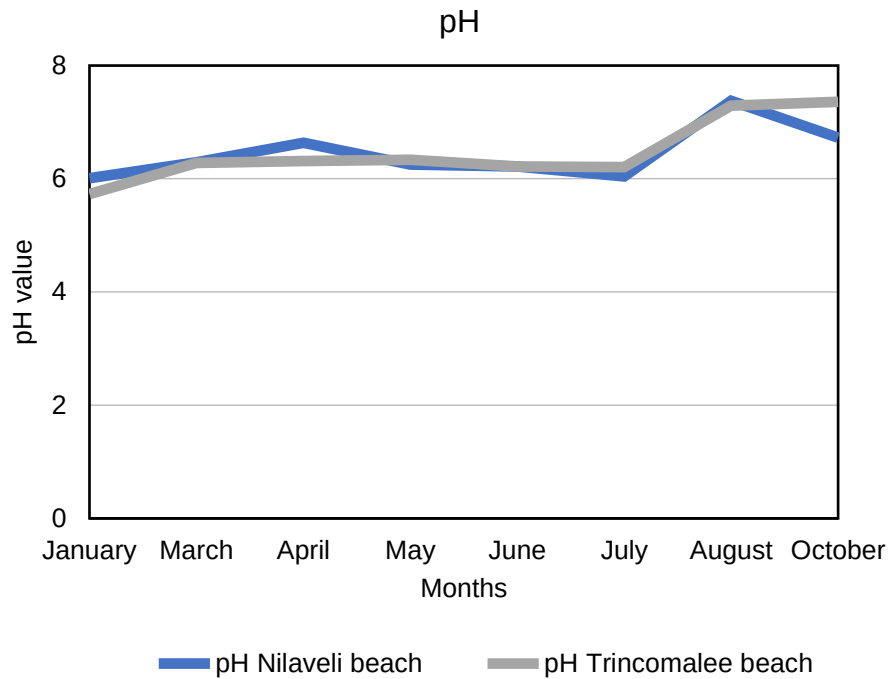
SALTINITY



Graph :07

The salinity range of ocean is 33-37 (ppt).According to our data average salinity is 34-35(ppt) it is acceptable. Nilaveli and Trincomalee beach side's salinity is approximately remains same in all time.

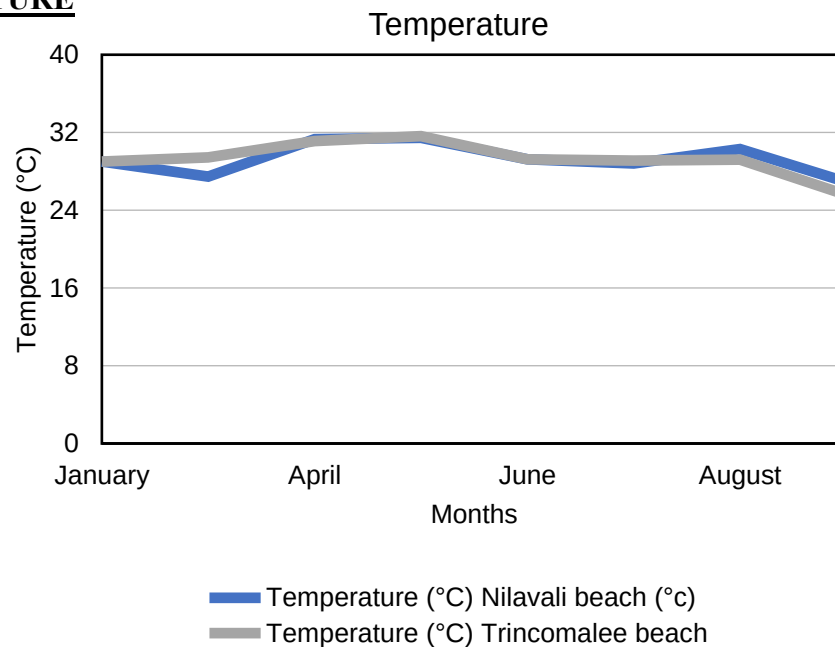
PH



Graph :08

pH is the key parameter to indicate water quality. The seawater pH range is 7.5-8.4. According to our data both beach side's average pH is 6.4 and it is looking acidity it is danger for corals. This may affect the surface of the coral because it's made by calcium carbonate. By comparing data Trincomalee's pH is slightly lower than nilaveli's pH.

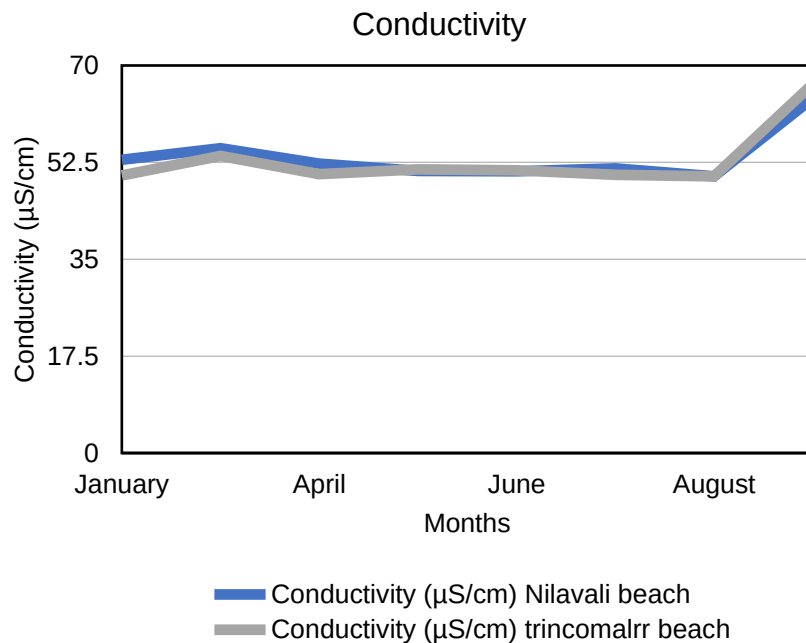
TEMPERATURE



Graph:09

The optimal temperature of coral reefs is 23 – 29(°C). Our average sewage temperature is 29.2 °C so temperature wise it is not going to harm the marine environment.

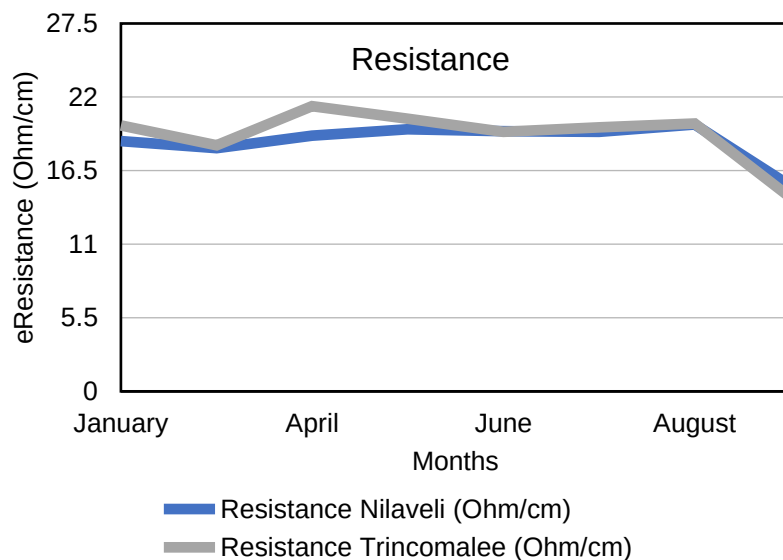
CONDUCTIVITY



Graph :10

The average conductivity of selected region's sewage is 53µS/cm which means it has not much of minerals or ions it is safe to mixed with sea.

RESISTANCE



Graph:11

High resistance indicate low ion concentration. low resistance indicate pollution due to dissolved ions, Our average resistance is 19 Ohm/cm. Overall, By comparing Trincomalee and nilaveli's data Trincomalee's sewage is little bit seems poor quality because Trincomalee have high population comparing to nilavelli.

8.Experts Insights

Coastal Conservation and Coastal Resource Management Department

Previous Projects:

Between 2010 and 2016, a large project focused on protecting coral reefs. Research indicated that coral reefs were increasingly being damaged by unchecked tourism and coastal activities. The reefs were assisted in recovery, but those improvements didn't stick.

Pigeon Island Management Plan

Pigeon Island is also segregated into areas used for bathing, tourism, and preserving its rich wildlife. Strict regulations exist to minimize the damage that individuals cause the coral reefs. Sustainable tourism guidelines are also helping to protect the marine ecosystems.

Destructive Erosion and Coastal Protection:

Erosion is a big problem in Thirukadalur, Nilevali, Muthur, Kinniya especially during heavy rains. Black stone barriers are a bulwark against this, but their effectiveness varies based on their location and the intensity of the waves. Marine projects aim to avoid sand loss and stabilize the shoreline.

Cleaning the Lakes and Rivers Natural]: Plastic Waste Management and Clean Sri Lanka:

To remove plastic waste, the Clean Sri Lanka initiative has conducted several beach cleanups across Trincomalee. "We're planning to bring these efforts to other coastal communities facing similar problems." Combating this issue, we are concentrating on two vehicles: Raising awareness among the public, and establishing stricter regulations regarding litter disposal.

Agricultural Biomass Waste and Boat Engine Oil:

8733It appears that castor oil, an agricultural waste, is ruining our beaches, according to reports. We're also seeing an increase in pollution due to oil from boat engines, which is affecting marine organisms and water quality. Experts say we need stricter rules for how waste is disposed of, as well as greener choices for fuel and farming practices.

Marine Environmental Protection Authority(MEPA)

Sample Testing:

We tested water samples for pH balance, salt content and dissolved oxygen content. These high acid levels imply that the ocean is becoming more acidic, and that is bad for coral health. Less dissolved oxygen can be a real game-changer for the range of life in the ocean.

Ocean Acidification And Coral Bleaching:

Coral bleaching is increasingly problematic as ocean temperatures rise. Exposure to harmful agents, including some chemicals in sunscreens, accelerates the coral's bleaching.

Initiatives to save coral include reducing chemical pollution and advocating for more eco-friendly products.

Xpress Pearl Disaster:

The Xpress Pearl is leaking dangerous materials into the ocean. This specially damaged already devastated coral reef ecosystems. We need to monitor closely over a long term to understand the full impact on biodiversity.”

Artificial Coral Videos:

We might be using artificial coral reefs to restore damaged marine environments. So far, the early results indicate that the artificial corals may help fish population numbers, which is a positive sign. However, further study is needed to determine if this technique consistently performed well over time.

SOLUTIONS

1. Implying Strict Rules

Coral reefs and coastal ecosystems play critical roles for biodiversity, fisheries, and coastal protection. Pollution from land-based activities, absence of regulations and lack of enforcement mechanisms are threatening coral reefs in the Trincomalee area. So tight regulations and following global best practices is all the more critical here.

Key Recommendations

- Legislative and Policy Writing

- o Pass legislation in support of coral reef protection with fines for pollution and habitat destruction

- o Make zoning laws to restrict damaging practices near coral reefs, like unmonitored tourism, fishery or coastal development.

- Water Quality Standards

- o Adopt water quality criteria based on coral health thresholds, as done under the U.S. Clean Water Act (CWA). These standards should control pollutants that negatively impact coral ecosystems, such as sediments, nutrients and chemical contaminants.

- o Regular monitoring of water quality in areas that discharge inputs to coral habitats for compliance with standards.

- Pollution Control Measures

- o Establish tighter controls on agricultural runoff through cuts in nitrogen and sediment discharge into coastal waters, as in Australia's Reef Protection Regulations.

- o Control point-source pollution by industries and sewage treatment plants with permits imposing limits on pollutants.

- MPAs (Marine Protected Areas)

- o Increase marine protected areas (MPAs) around key coral sites in Trincomalee to safeguard coral, including banning harmful practices.

- o Implementation with strict monitoring and penalties for any violations.

- Community Education and Involvement

- o Raise awareness among local communities on the significance of coral reefs and consequences of pollution

- o Suggest citizen involvement in monitoring programs to report illegal activities.

- Research and Gathering of Information

- Help conduct research on pollutants affecting corals in the region, in partnership with universities and NGOs.

- o Apply scientific results to inform updated regulations and new enforcement mechanisms.

- Intersectoral Coordination

- o Form a task force consisting of government, NGOs, and local communities to streamline the activities to protect coral reefs.

- o Encourage terrestrial regulators to work with marine conservationists to tackle land-based sources of pollution.

2. Proposals for Land Use Zoning in Trincomalee

- Create Marine Protected Areas (MPAs)

- o Establish MPAs in zones where the coral reefs and coastal land are ecologically sensitive.

- o Designate no-take zones, where all extractive activities, including fishing and mining, are prohibited, so that ecosystems can rebound and prosper as in Australia's Great Barrier Reef Marine Park zoning.

- Reassign Land Zoning Respectively

- o Globally designate coastal and marine areas as zones for different activities:

- Conservation Zones: Land areas that are strictly protected to conserve biodiversity.

- Buffer Zone: Develop buffer zones around important habitats to minimize the effects of terrestrial pollution. These zones may shield reefs from agricultural run-off or industrial waste.
- Eco Tourism Zones: Areas where eco-tourism will happen under strict rules so that they do not affect the environment.
- Fishing Zone: where sustainable fishing practices are permitted.
- Development zones: Only some areas are designated for commercial coastal development under stringent environmental guidelines.

- High-Impact Activities Control

- o Restrict activities like coral mining, waste disposal, and destructive fishing practices (bottom trawling) in sensitive areas.

- o Control development along shorelines with beach setbacks and green building to limit coastal erosion.

- Conflict Management

- o Deploying zoning to segregate conflicting activities (such as, tourism and fishing), so that user conflicts are minimum and sustainable use of resource is ensured;

- Monitoring and Enforcement

- o Build a stringent enforcement mechanism to check violations of zoning laws.

Deploy patrols and surveillance systems, as they do in the Florida Keys National Marine Sanctuary, to ensure that rules are policed effectively.

- Community Involvement

Engaging the local communities in the zoning process by taking into account their knowledge and ensuring their livelihood is not compromised.

- o Offer alternative livelihoods (e.g., eco-tourism jobs) for communities that were previously dependent on restricted access areas.

3.Support Planting of Artificial Corals

- **Reef Restoration** — Introduce artificial coral programs in degraded sites, using limestone, eco-friendly concrete. This can restore habitats and improve biodiversity.
- **Community Involvement:** Engage and train local communities in coral gardening techniques to create ownership and involvement. This can create employment and create sustainable livelihoods.
- **International Highlight:** The Maldives have used artificial reefs to help restore damaged ecosystems. Study these models, and adapt them to local conditions.

Research and Development: Ensure that the above methods are effective and sustainable by funding the research of the continuous improvement of the techniques used in the artificial coral planting.

4. Increase Public Awareness

- **Educational Campaigns:** Run awareness campaigns in educational institutions, communities, and via media to inform about the significance of coral reefs and their contribution towards coastal protection. This may comprise workshops, documentaries, and social media campaigns.
- **Behavioral Change Programs:** promote sustainable practices such as reducing plastic use, responsible waste disposal and eco-friendly fishing methods. Collaborate with local businesses to encourage sustainable tourism practices.
- **Cultural Integration** — Use traditional knowledge and cultural values in awareness programs. This might make conservation more personal and effective.
- **Influencers Partnership:** Collaborate with popular influencers or celebrities within the community to ensure the message gets spread with major reach.

Step 1. Organize Beach Cleanup Programs

- **Regular Events:** Organize regular Beach Clean-up with community members, visitors and NGOs. This will also encourage community involvement and raise awareness about pollution.
- **Data Collecting Events:** Leverage those events to collect data for better waste management strategies. Draft reports and recommendations based on the above data.
- **Partnerships:** Partner with international organizations, such as the Ocean Conservancy's International Coastal Cleanup program, to pool resources and knowledge.

- **Education and Outreach:** Educate participants about pollution and encourage sustainable practices during cleanup events.

5. Shading Reefs

- **Temporary Shading Structures:** During extreme heat events, use shading systems over priority reef areas to prevent coral bleaching. This can be an interim measure until sustainable solutions can be found.
- **Pilot Projects:** Evaluate shading technologies in Australia's Great Barrier Reef as a model for implementation. Evaluate their efficacy and adjust them for local contexts.
- **Research and Development:** Keep your eyes focused on advancing your achievement to improve the technology of shading.
- **Community Involvement:** Utilizing local communities in the deployment and maintenance of shading structures encourages participation and support.

5. Include both Monitoring and Reporting

- **Real-Time Data Collection:** Use technology such as underwater drones and sensors to track the health of reefs. It can offer early warning signs of stressors such as bleaching or pollution.
- **Citizen Science Programs:** Enlist local divers and fishermen to report alterations in reef conditions. This may really inform reef health and show where we need to intervene.”
- **Transparent Reporting Systems:** Public dashboards or reports on findings shared with stakeholders. This can help establish trust and guarantee accountability in conservation efforts.
- **Use of Evidence-Based Decision-Making:** Collect data to support policy decisions and conservation strategies.

6. Address Unmanageable Plans

- **Feasibility studies:** Prioritize projects to maximize cost effectiveness and outcomes from environmental impact assessments. It also helps ensure plans are grounded in reality and realistically able to achieve the outcome with the resources available.
- **Stakeholder Engagement:** Engage local communities, governmental bodies, and experts throughout the planning process to ensure that plans are tailored to local needs and are realistic.
- **Flexible Planning:** Plans should change as conditions change and new information enters.

- Collaborative Governance: Bring government, NGOs and community partners around the table to enable supportive and implemented plans.

7.Explore Probiotic Treatments

- Coral Microbiome: Invest in research to develop probiotic treatments that improve coral resilience under bleaching and other stressors. This may include partnerships with universities or research institutions.
- Pilot Projects: Partner with research institutions currently trialing probiotics for corals. Track the efficacy of these treatments and ramp up successful interventions.
- Regulatory Frameworks: Establish protocols for the use of probiotics in coral conservation to avoid unwanted effects.
- Community education: Educate the community about the potential benefits and risks associated with probiotic treatments and address any misconceptions to promote acceptance and participation.

8.Restore Coral Reefs

- Rehabilitation Approaches: Employ measures like coral transplantation or larval reseedling to restore damaged reefs. This can support biodiversity and build resilience in ecosystems.
- Continued Restoration Resource — International grants or carbon trading to fund long-term restoration. Earmark funding that is dependable and sustainable.
- Community Involvement: Mobilize local communities to participate in restoration efforts to promote ownership and sustainability.
- Monitoring and Evaluation: Regularly monitor restoration sites to assess restoration outcomes and improve techniques.

9.Maximize the Reach of Public Information

Even though it may be an effort, knowledge often lands short of action:

- Localized Messaging: Messages should be tailored to the local cultural and economic context. Use storytelling plus local examples to make conservation relatable

- **Community Heroes:** Find leaders in the community who will champion protecting the reef. These champions can further amplify the message and spur action on a broad scale.
- **Behavioral Incentives:** Encourage sustainable behaviors through rewards, like discounts for environmentally conscious decisions or public acknowledgment for conservation activities.
- **Feedback Mechanisms:** Create feedback loops to know why awareness is not translating into action and adjust strategies accordingly.

10.Climate Adaptation Plans:

- Embed coral reef conservation in broader climate adaptation strategies — as promoted at global meetings such as COP27. That may include preparing for sea-level rise, a greater number of storm events or ocean acidification.
- Establish resilience plans for coastal communities that rely on coral reefs for livelihoods.

11.Reduce Land-Based Pollution:

- Upgrade wastewater treatment facilities and encourage sustainable agriculture so that less fertilizer ends up in coastal waters. This includes policies on the use of fertilizers and waste management.
- Advocate for sustainable practices from industries within coastal regions.

12.Eco-Tourism Development:

- Encourage eco-tourism projects that protect natural resources while bringing in revenue. Such activities may include training local guides, creating sustainable tour packages, and advocating responsible tourism behaviors.
- Make sure local communities share equitably in the benefits of tourism, to encourage support for conservation.

Enlightened Recommendations with Worldwide Illustrations

What other countries are doing, respectively point by point, to address pollution on corals and coastlines

1. Enforce Strict Rules

- **United States:** The U.S. invokes the Coral Reef Conservation Act to safeguard coral ecosystems from climate change, pollution and overfishing. The law also emphasizes sustainable

uses, research and the benefits to communities while imposing penalties for violations such as illegal fishing and destroying coral.

Australia: The Great Barrier Reef Marine Park Authority regulates the reef's Marine Protected Area (MPA) through strict management plans. They are heavily penalized for operating in sensitive areas, overfishing, and pollution, among other activities. The authority also carries out regular patrols to ensure compliance.

- International Cooperation: The International Coral Reef Initiative (ICRI) helps strengthen the legal framework for coral worldwide. It helps fund regional workshops, and aids countries in creating national policies to protect reefs.
- France: France has strict policies in place for its overseas territories, including New Caledonia, where reefs are protected by national biodiversity laws.

2. Implement Zoning

- Australia: The zoning plan for the Great Barrier Reef divides the reef system into a number of zones for conservation, tourism, and fishing. More than a third of the reef is designated a “no-take” zone, in which fishing and extraction are banned.
- Philippines: Zoning in MPAs is implemented by local governments, demarcating areas for tourism, fishing and conservation. This makes community-managed marine sanctuaries a key feature of the approach.
- United States: In MPAs, NOAA uses zonation to regulate for activities such as fishing and coastal development. For example, Florida's coral reefs are managed using a zoning scheme that allows for a balance between ecological protection and economic activities.
- Japan: Okinawa enforces zoning laws that limit types of industrial activities near coral reefs and encourage eco-tourism in certain regions.

Contribute to the Artificial Reef Construction

- Maldives: Eco-friendly materials have been used extensively as artificial reefs for coral ecosystem rehabilitation. Such structures also draw in marine life and promote biodiversity.
- Thailand: Underwater artificial reef projects where concrete blocks and steel frames are used to create habitat for corals and other marine organisms. One of the main parts of these efforts is community involvement.
- United States: Large-scale out planting efforts transplant corals back onto degraded reefs with the expectation that restoration will accelerate (NOAA's Coral Reef Conservation Program).

And Indonesia: Local communities in partnership with NGOs pioneering methods such as Bio rock technology, which uses low-voltage electricity to grow the coral planting artificial corals.

3. Increase Public Awareness

- Australia: Public awareness campaigns as part of the Reef 2050 Long-Term Sustainability Plan inform citizens about sustainable behavior, such as avoiding plastic waste, and saving water. Initiatives focus on schools and community businesses.
- United States: The EPA holds campaigns for educating communities on how their choices — like reducing runoff or steering clear of harmful sunscreens — can help coral reefs get the care they need.
- Philippines: Grassroot organizations organize workshops on sustainable fishing practice and how coral reefs are vital for local livelihoods. Traditional ecological knowledge is frequently integrated into these programs.
- International Year of the Reef (IYOR): In order to raise global awareness on the importance of coral reef conservation, ICRI declared 2018 as IYOR and built events, educational materials, and campaigns across social media which were implemented across member countries like Jamaica, Japan and Sweden.

4. Organize Beach Cleanup Programs

- United States: States with coasts, such as Florida, frequently hold beach cleanups with volunteers, NGOs and local governments. The data collected during these events is used to track the sources of marine debris.
- Japan: Community-led clean-ups of the beach are common in Okinawa, and local schools are involved in keeping communities clean, while removing debris from beaches close to budding coral microbes. Many of these events include educational sessions relating to marine conservation.
- Maldives: NGOs have partnered with resorts to plan regular beach cleanups as a part of eco-tourism initiatives. These programs are open to tourists, and visitors are encouraged to join in while they are there.
- India: Tamil Nadu and other coastal states hold cleanup drives under the Swachh Bharat Abhiyan campaign to combat marine plastic pollution.

5. Combine Monitoring and Reporting

- United States: NOAA’s Coral Reef Monitoring Program employs emerging technologies such as underwater drones and satellite imaging to monitor reef health. Citizen science projects also enlist the help of divers and fishermen to submit observations via mobile apps.

You are focused on data up to October 2023• Australia: The Great Barrier Reef Marine Park Authority carries out comprehensive monitoring using the Reef Integrated Monitoring and Reporting Program (RIMREP) which gathers information on water quality, biodiversity and anthropogenic stressors affecting the reef

- Global Coral Reef Monitoring Network (GCRMN): GCRMN is a regional process that operates under ICRI’s umbrella providing standardized instruments for measuring reef health to regions including the Caribbean, East Asia, and Pacific. About the Speaker: In this episode, we have Andrea Onofri, the Conversation Designer at Cognizant who contributes to the development of Global Standards for Data Collection (GSDC) to ensure an universally consistent way for data collection.

- Philippines: Community-based monitoring programs teach local fishers to document changes in reef conditions with simple tools, such as underwater cameras or logbooks.

6. Restore Coral Reefs

- United States: NOAA’s Coral Reef Conservation Program focuses on deploying nursery-grown corals onto degraded reefs as part of large-scale restoration projects. That will make others begin to grow faster, using techniques such as micro fragmentation.
- Australia: The Reef Restoration and Adaptation Program (RRAP) has been responsible for designing and testing various restoration methods, and some of the most innovative measures, such as larval reseedling and genetic engineering methods aimed at increasing coral resilience to climate change effects, have been developed within that program.
- Maldives: Integrated into eco-tourism initiatives where coral restoration is a supporting feature, resorts fund projects, including coral gardening or out planting activities with tourists to participate in.
- Indonesia: Community-based mangrove management through partnerships with NGOS to help restore mangroves near coral reefs, as mangroves serve as natural filters for sedimentation, which may damage coral.

Summary Table of Global Examples

Recommendation	Country Examples
Enforce Strict Rules	U.S., Australia (Great Barrier Reef), France (New Caledonia), ICRI collaboration
Implement Zoning	Australia (zoning plans), Philippines (community MPAs), Japan (Okinawa)
Promote Artificial Coral Planting	Maldives (eco-materials), Thailand (concrete blocks), Indonesia (Biorock)
Increase Public Awareness	Australia (<i>Reef 2050</i> campaigns), U.S., Philippines (grassroots workshops)
Organize Beach Cleanup Programs	U.S., Japan (Okinawa), Maldives (resort partnerships), India
Integrate Monitoring & Reporting	U.S. (NOAA tech), Australia (<i>RIMREP</i>), GCRMN
Restore Coral Reefs	U.S., Australia (<i>RRAP</i>), Maldives (tourist-funded projects), Indonesia

These global examples demonstrate how different countries effectively implement strategies tailored to their specific challenges while contributing valuable lessons for Trincomalee's efforts in addressing coral and coastal

CONCLUSION

Our study highlights the urgent need for effective conservation strategies to protect coral reefs and coastal areas. Notably, a **zoning plan has not yet been implemented in Trincomalee's Pigeon Island**, which could provide significant protection if enforced. While seasonal coastal erosion affects our study area, placing barriers to mitigate erosion on one side often shifts the impact to another, requiring a more comprehensive approach.

Currently, many government organizations have implemented artificial coral planting and other restoration efforts. If the government prioritizes long-term conservation strategies, including proper zoning and sustainable management, we can ensure better protection for coral reefs. In contrast, private organizations primarily focus on short-term initiatives, which may not provide lasting benefits.

Public awareness and protection efforts, such as those under the **Clean Sri Lanka** program, serve as positive examples, but their involvement remains limited. Strengthening government driven programs and increasing public engagement are crucial for the long-term sustainability of coral and coastal ecosystems.

CHALLENGES AND LIMITATIONS

In our project, we faced number of challenges/limitations that impacted our ability to effectively identify and analyze data:

1) Few Facilities for Widespread Laboratory Testing

After laboratory tests, we aimed to measure coral samples, soil, and ocean water to gain real-time data and experiences. However, we could not perform these analyses due to lack of facilities and high cost of testing.

2) Impact on Academic Schedule

We had to miss lectures every day to collect data from the official departments, which in itself was hard to manage. It was difficult to balance both of field visits and coursework.

3) No Data from the Coastal Conservation and Management Department

The Coastal Conservation and Management Department, which we visited, said they do not have records on coastal erosion. Their strategy is to wait until someone complains to act. Because there is no system of data collection in place, we could not get any records of past or present erosion. We were told to get such data from their headquarters in Colombo. But because of high travel costs and time constraints, we decided to request information by email, but got no reply. Certain reports, for instance, required payment to be obtained.

4)Data Limitations from MEPA

The Marine Environment Protection Authority (MEPA) revealed that Sri Lanka struggles with data collection and analysis on marine pollution. They do not keep detailed records, and earlier data is not publicly accessible. Moreover, they said that sharing unpublished data had the potential to result in misinterpretation or erroneous conclusions.

CONTRIBUTION OF MEMBERS

M.Miruthula	• Data Analysis • Objectives • Background of Study • Methodology • Field Observation • Designing the Report
Amirthalingam.L	• Problem Statement • Presentation Slides • Designing the Report
Archchana.K	• Introduction • Literature Review 2 • References
Kogulasriram.J	• Literature Review 1 • Presentation Slides • Data Analysis • References
Divigsharan.V	• Literature Review 3 • Solutions • Taking Photos • References
Girithigan.J	• Solutions • Timeline • References



REFERENCES

[1]S. Mehvar, A. Dastgheib, J. Bamunawala, M. Wickramanayake, and R. Ranasinghe, "Quantitative assessment of the environmental risk due to climate change-driven coastline recession: A case study in Trincomalee coastal area, Sri Lanka," *Department of Water Engineering and Management, University of Twente, Enschede, The Netherlands; Department of Water Science and Engineering, IHE Delft Institute for Water Education, Delft, The Netherlands; Coast Conservation and Coastal Resource Management Department, Colombo, Sri Lanka; Harbour, Coastal and Offshore Engineering, Deltares, Delft, The Netherlands.*

[2]**Coast Conservation and Coastal Resource Management Department, Sri Lanka Revised Coastal Zone and Coastal Resource Management Plan 2024 - 2029**, Colombo, Sri Lanka: CC&CRMD, 2024. [Online]. Available: https://www.coastal.gov.lk/images/pdf/CZMP_24-29/CZCRMP_2024_PC_ENG.pdf. [Accessed: Mar. 04, 2025].

[3]"Marine spatial planning," *Wikipedia, The Free Encyclopedia*. [Online]. Available: https://en.wikipedia.org/wiki/Marine_spatial_planning. [Accessed: Mar. 5, 2025].

[4]L. Bo-yu, "Thailand exempts land tax to boost blue carbon in mangrove forests," *Reccessary*, Dec. 23, 2024. [Online]. Available: <https://www.reccessary.com/en/news/th-regulation/thailand-exempts-land-tax-mangrove-carbon-credits>. [Accessed: Mar. 5, 2025].

[6]"**Indonesia drafts regulation to boost protection value of mangroves**," *Antara News*, Feb. 29, 2024. [Online]. Available: <https://en.antaranews.com/news/338350/indonesia-drafts-regulation-to-boost-protection-value-of-mangroves>. [Accessed: Mar. 5, 2025].

- ✓ [Sewage pollution: mitigation is key for coral reef stewardship - PubMed](#)
- ✓ <https://www.sciencedirect.com/book/9780128118368/the-indian-nitrogen-assessment>
- ✓ <https://data.worldbank.org/indicator/AG.CON.FERT.ZS>
- ✓ "climate change and ocean ecosystem : protecting coral reefs in Sri Lanka", SLYCAN Trust, [online] Available: <https://slycantrust.org> , [Accessed : 05 March 2025]
- ✓ Michael BW, Nadine H et.al, *The Value of Reefs for Protecting the Most Vulnerable Populations in the Dominican Republic, Jamaica and Grenada* (InsuResilience Global Partnership, 2020)
- ✓ The World Bank, *Sri Lanka: Managing Coastal Natural Wealth* (World Bank Group, 2017)
- ✓ Mallon J, *Coral Reef Conservation and the Role of Blue Carbon* (Operation Wallacea, 2018)
- ✓ Herr D, Landis E, *Coastal blue carbon ecosystems: Opportunities for Nationally Determined Contributions Policy Brief* (IUCN, 2016)
- ✓ Berg H, Ohman M et.al, 'Environmental Economics of Coral Reef Destruction in Sri Lanka' [1998] 27(8) Royal Swedish Academy of Sciences
- ✓ "90 percent of Sri Lanka's are dead", International Collective in Support of Fish works, [online] Available: <https://www.eurasiareview.com/11062019-90-percent-of-sri-lankas-corals-are-dead/> [accessed: 05 March 2025]
- ✓ "5 coral reefs of Sri Lanka : Current status and Resource Management by Arjan Rajasuriya", fao.org [online] Available: <https://openknowledge.fao.org> [Accessed: 06 March 2025]
- ✓ "In a former conflict zone in Sri Lanka, a world rich in corals thrives", MONGABAY, [online] Available: <http://news.mongabay.com> , [Accessed : 05 March 2025]
- ✓ "Reviving Sri Lanka's Coral Reefs", , [online] Available: <https://biodiversitysrilanka.org>, [Accessed : 06 March 2025]

- ✓ "Anthropogenic nitrogen pollution threats and challenges to the health and challenges to the health of south Asian coral reefs", *Frontiers*, [online] Available: <https://frontiersin.org>, [Accessed : 06 March 2025]
- ✓ [1] United Nations Environment Programme, "International Coral Reef Initiative," [Online]. Available: <https://www.unep.org/explore-topics/oceans-seas/what-we-do/working-regional-seas/partners/international-coral-reef>. [Accessed: Mar. 4, 2025].
- ✓ [2] National Oceanic and Atmospheric Administration (NOAA), "Protecting coral reefs: Laws and treaties," [Online]. Available: https://oceanservice.noaa.gov/education/tutorial_corals/coral13_protecting-coral-law.html. [Accessed: Mar. 4, 2025].
- ✓ [3] Coral Digest, "Reef regulations," [Online]. Available: <http://coraldigest.org/index.php/ReefRegulations>. [Accessed: Mar. 4, 2025].
- ✓ [4] National Oceanic and Atmospheric Administration (NOAA), "Protecting coral reefs," [Online]. Available: https://oceanservice.noaa.gov/education/tutorial_corals/coral11_protecting.html. [Accessed: Mar. 4, 2025].
- ✓ [5] South Carolina Journal of International Law and Business, "International coral reef initiatives," [Online]. Available: <https://scholarcommons.sc.edu/cgi/viewcontent.cgi?article=1213&context=scjilb>. [Accessed: Mar. 4, 2025].
- ✓ [6] United Nations, "International Coral Reef Initiative," [Online]. Available: <https://sdgs.un.org/partnerships/international-coral-reef-initiative>. [Accessed: Mar. 4, 2025].
- ✓ [7] U.S. Environmental Protection Agency (EPA), "What EPA is doing to protect coral reefs," [Online]. Available: <https://www.epa.gov/coral-reefs/what-epa-doing-protect-coral-reefs>. [Accessed: Mar. 4, 2025].
- ✓ [8] U.S. Environmental Protection Agency (EPA), "What you can do to help protect coral reefs," [Online]. Available: <https://www.epa.gov/coral-reefs/what-you-can-do-help-protect-coral-reefs>. [Accessed: Mar. 4, 2025].